

DCDC Converters

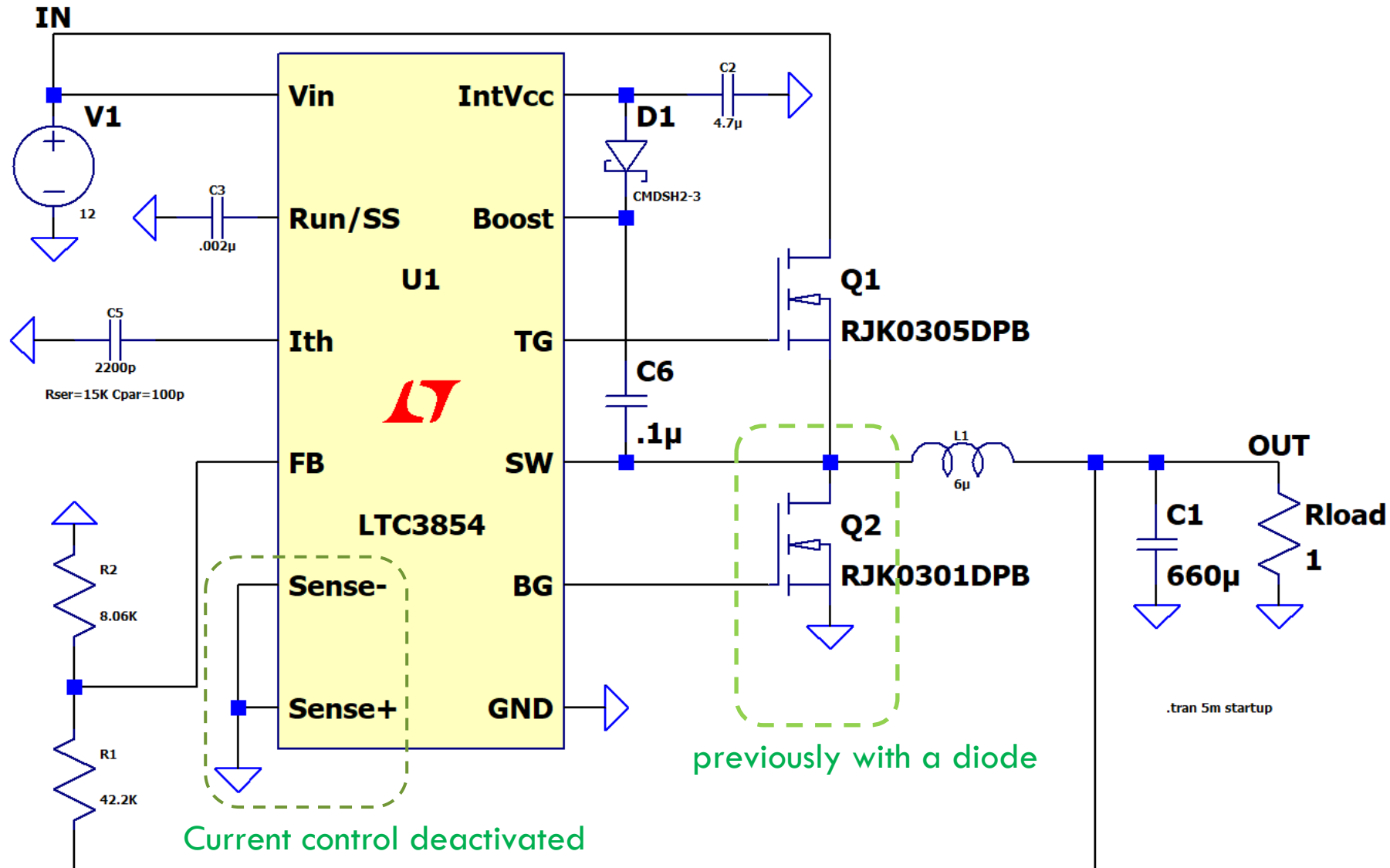
DCDC CONVERTERS – APPLICATION EXAMPLE

- Design a Buck DCDC Converter which converts an input voltage of 10 V down to 5 V without any feedback control
- What happens if the duty cycle is changed?
- How does the output voltage look like with and without the output filter (decoupling) capacitor?
- Implement a feedback circuit which adjusts the output voltage, simulate the inductor current and the output voltage with and without the filter capacitor



DCDC CONVERTERS – APPLICATION EXAMPLE

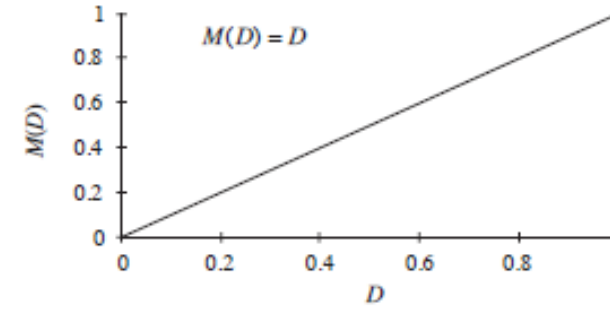
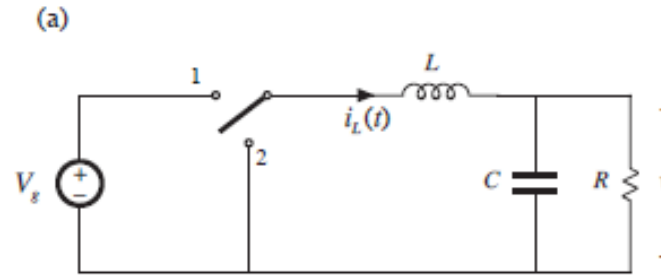
Design a Buck DCDC Converter which converts an input voltage of 12 V down to 5 V with a commercially available IC



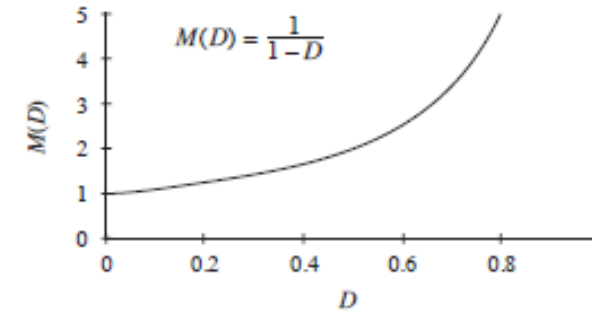
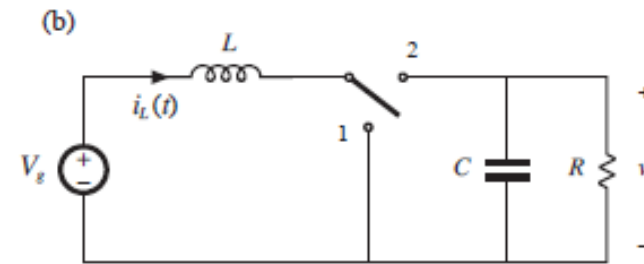
DCDC CONVERTERS

Three basic converters

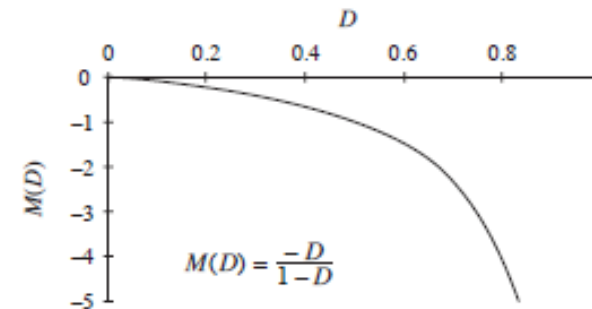
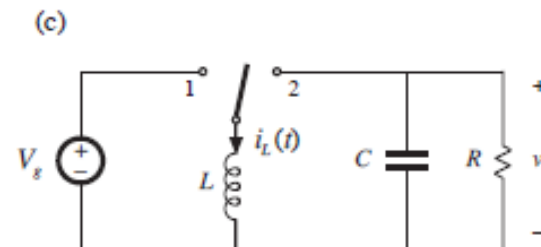
Buck



Boost



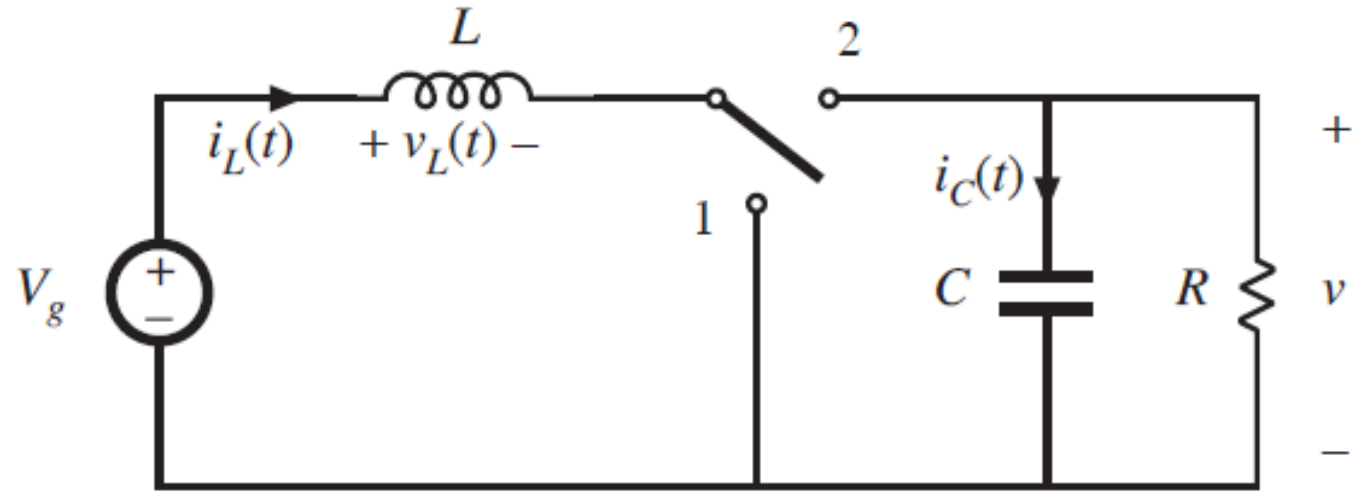
Buck-boost



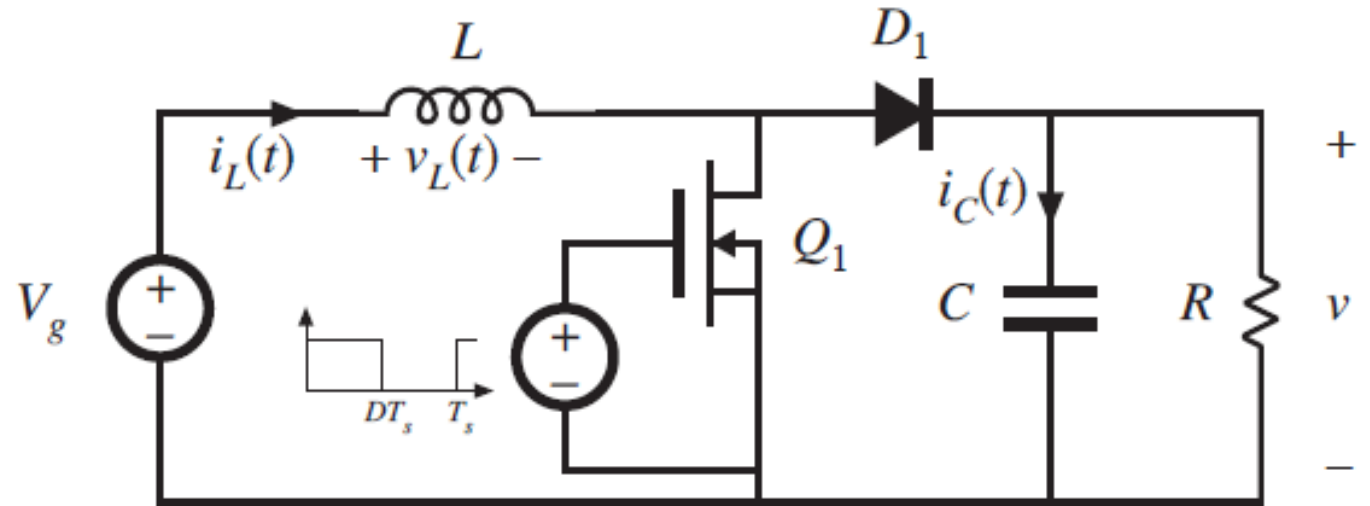
DCDC CONVERTERS

Boost Converter Analysis

*Boost converter
with ideal switch*

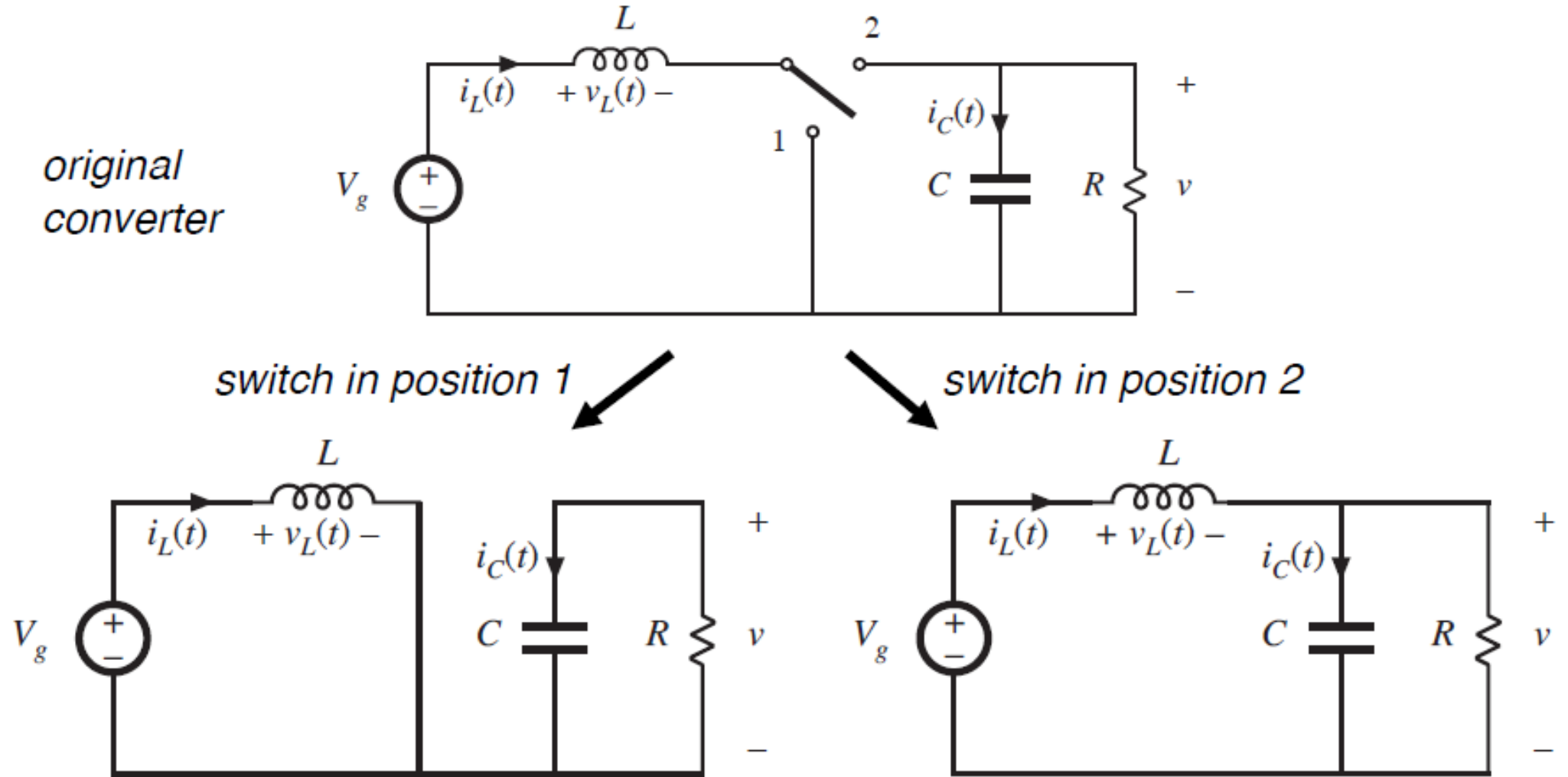


*Realization using
power MOSFET
and diode*



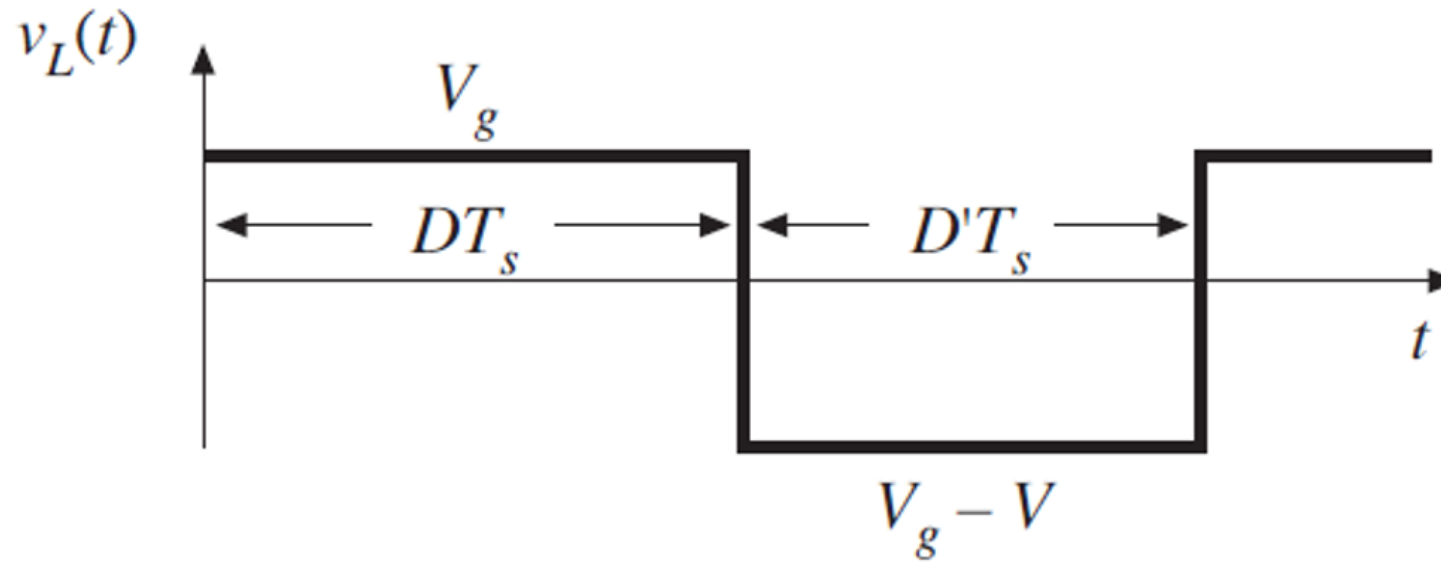
DCDC CONVERTERS

Boost Converter Analysis



DCDC CONVERTERS

Boost Converter Analysis

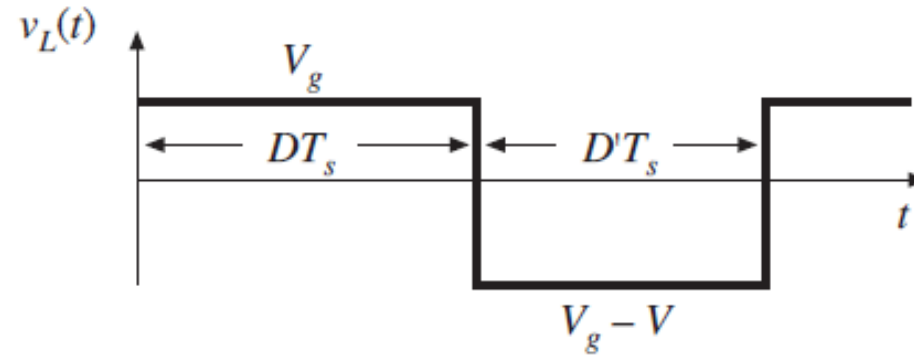


DCDC CONVERTERS

Boost Converter Analysis

Net volt-seconds applied to inductor over one switching period:

$$\int_0^{T_s} v_L(t) dt = (V_g) DT_s + (V_g - V) D'T_s$$



Equate to zero and collect terms:

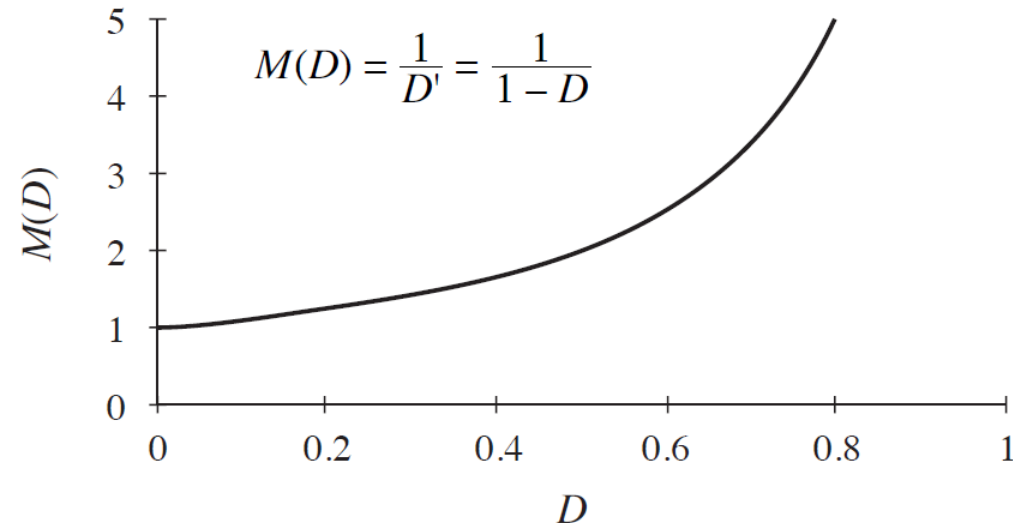
$$V_g (D + D') - V D' = 0$$

Solve for V:

$$V = \frac{V_g}{D'}$$

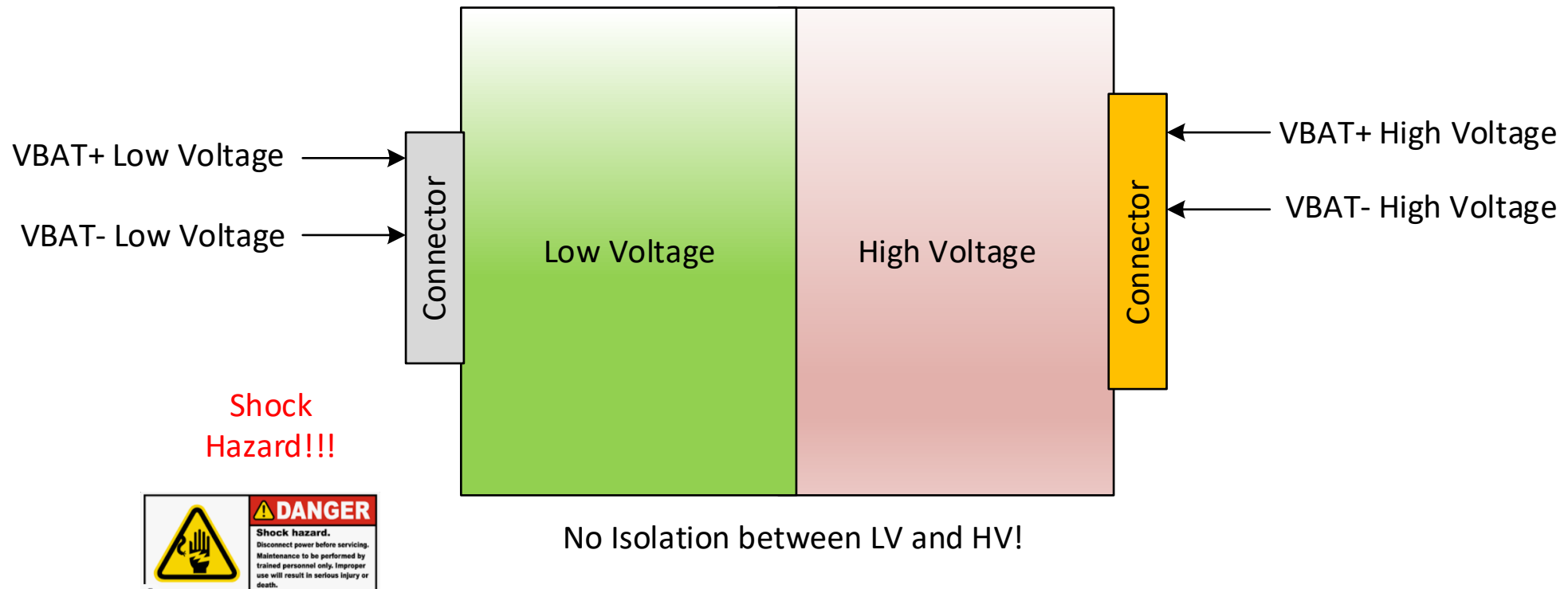
The voltage conversion ratio is therefore

$$M(D) = \frac{V}{V_g} = \frac{1}{D'} = \frac{1}{1 - D}$$



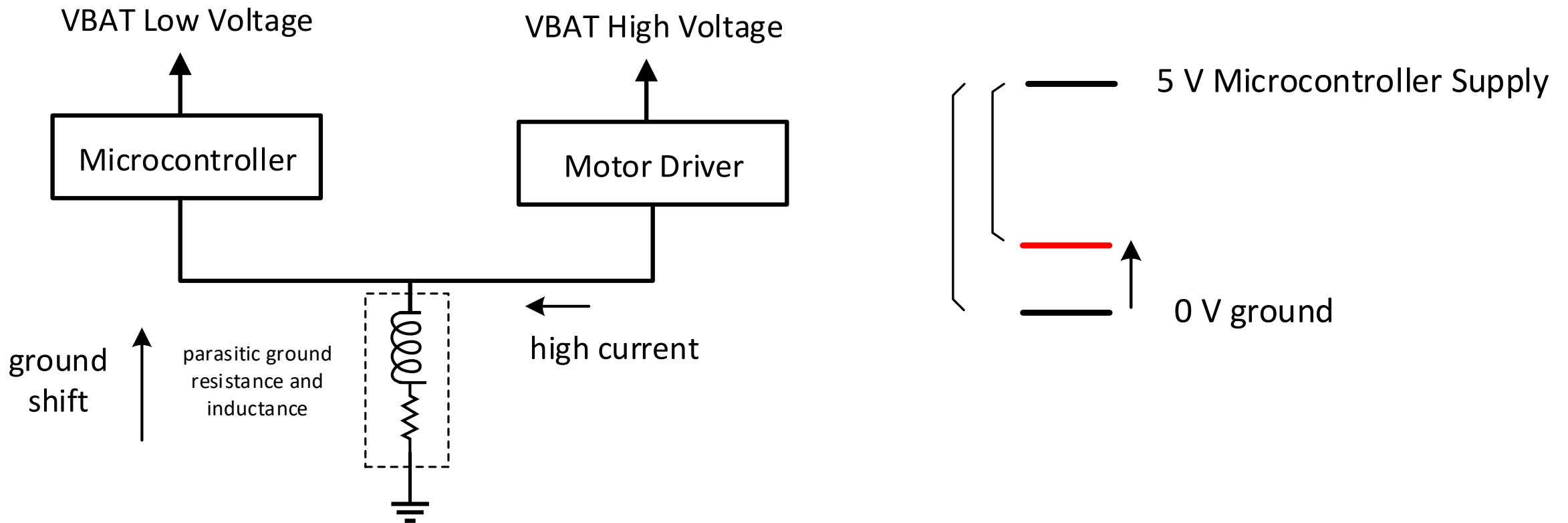
ISOLATED DCDC CONVERTERS

- A vehicle can have more than two batteries with different voltages, e.g. 48V Mild Hybrid Systems or 400V High Voltage Battery
- This might lead to high voltage electric shock at the low voltage connector if, for example, a technician touches the connector pins during maintenance
- Moreover, the high currents in the high voltage domain might lead to a ground shift in the LV domain. If there is a safety critical component like a microcontroller in the LV domain, the microcontroller will not work properly due to ground shift.



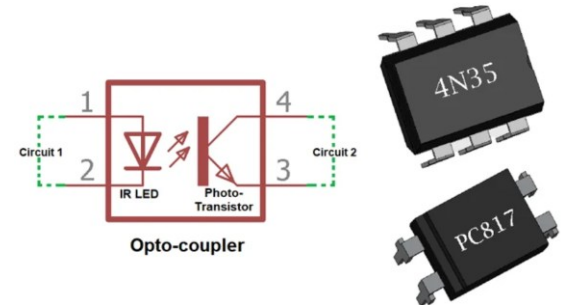
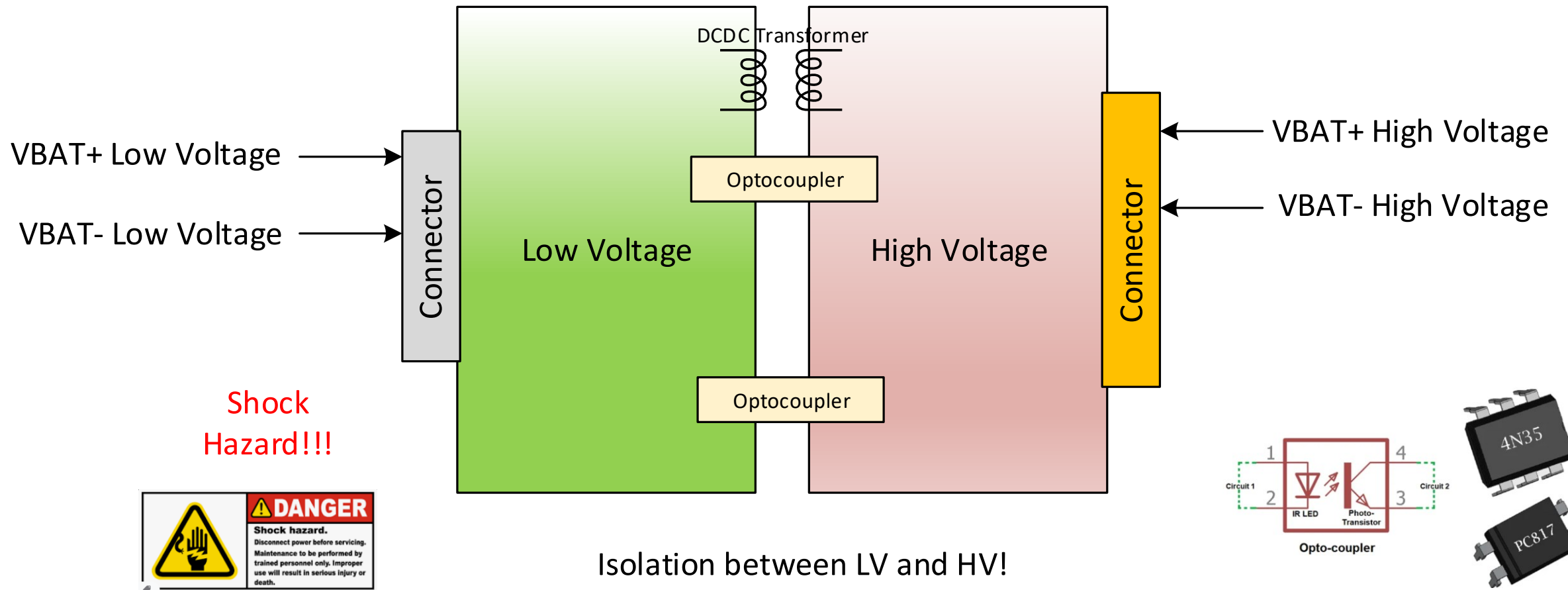
ISOLATED DCDC CONVERTERS

- Moreover, the high currents in the high voltage domain (noise) might lead to a ground shift in the LV domain. If there is a safety critical component like a microcontroller in the LV domain, the microcontroller will not work properly due to ground shift.



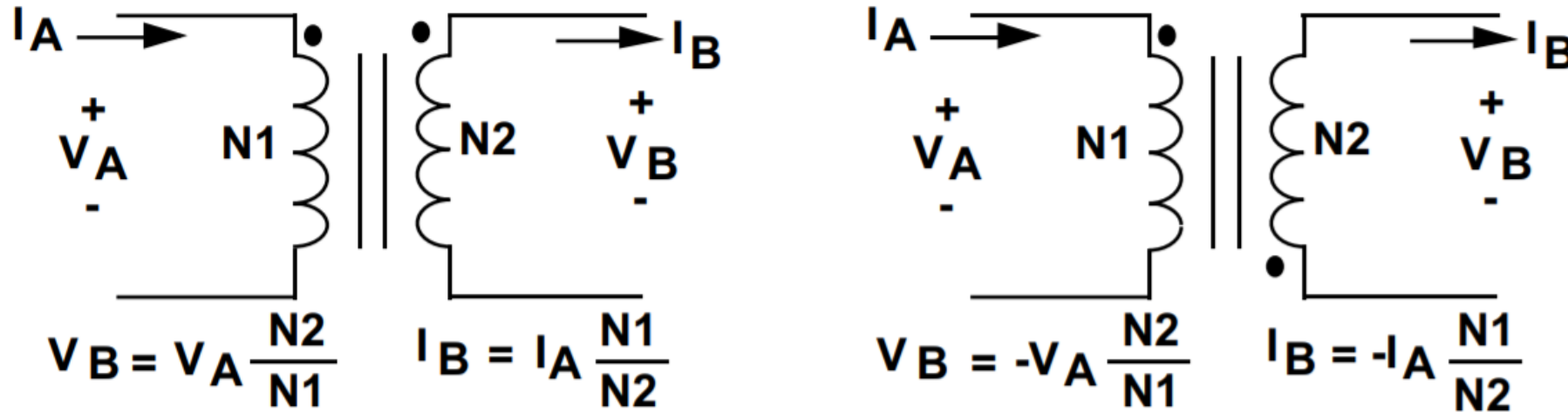
ISOLATED DCDC CONVERTERS

- This can be mitigated by isolating the LV and HV domains. The energy transfer between LV and HV can be realized by using isolated DCDC converters. Digital and analog signals can be transferred between LV and HV by using isolators (optical, capacitive or inductive)



FLYBACK DCDC CONVERTERS - TRANSFORMERS

A transformer is a device that has two or more magnetically-coupled windings. The basic operation is shown



The action of a transformer is such that a time-varying (AC) voltage or current is transformed to a higher or lower value, as set by the transformer turns ratio. The transformer does not add power, so it follows that the power ($V \times I$) on either side must be constant. That is the reason that the winding with more turns has higher voltage but lower current, while the winding with less turns has lower voltage but higher current. The dot on a transformer winding identifies its polarity with respect to another winding and reversing the dot results in inverting the polarity.

FLYBACK TRANSFORMERS

In a flyback converter, the magnetic component is often called a flyback transformer, but it actually behaves like a coupled inductor. Unlike a normal transformer, which transfers energy directly from the primary to the secondary while both conduct simultaneously, the flyback stores energy in the magnetic core during the switch-on period and releases it to the output during the switch-off period.

This means the core must include an air gap to allow energy storage.

Therefore, a flyback transformer functions as both an isolation device and an energy storage element, combining the roles of a transformer and an inductor in one component.

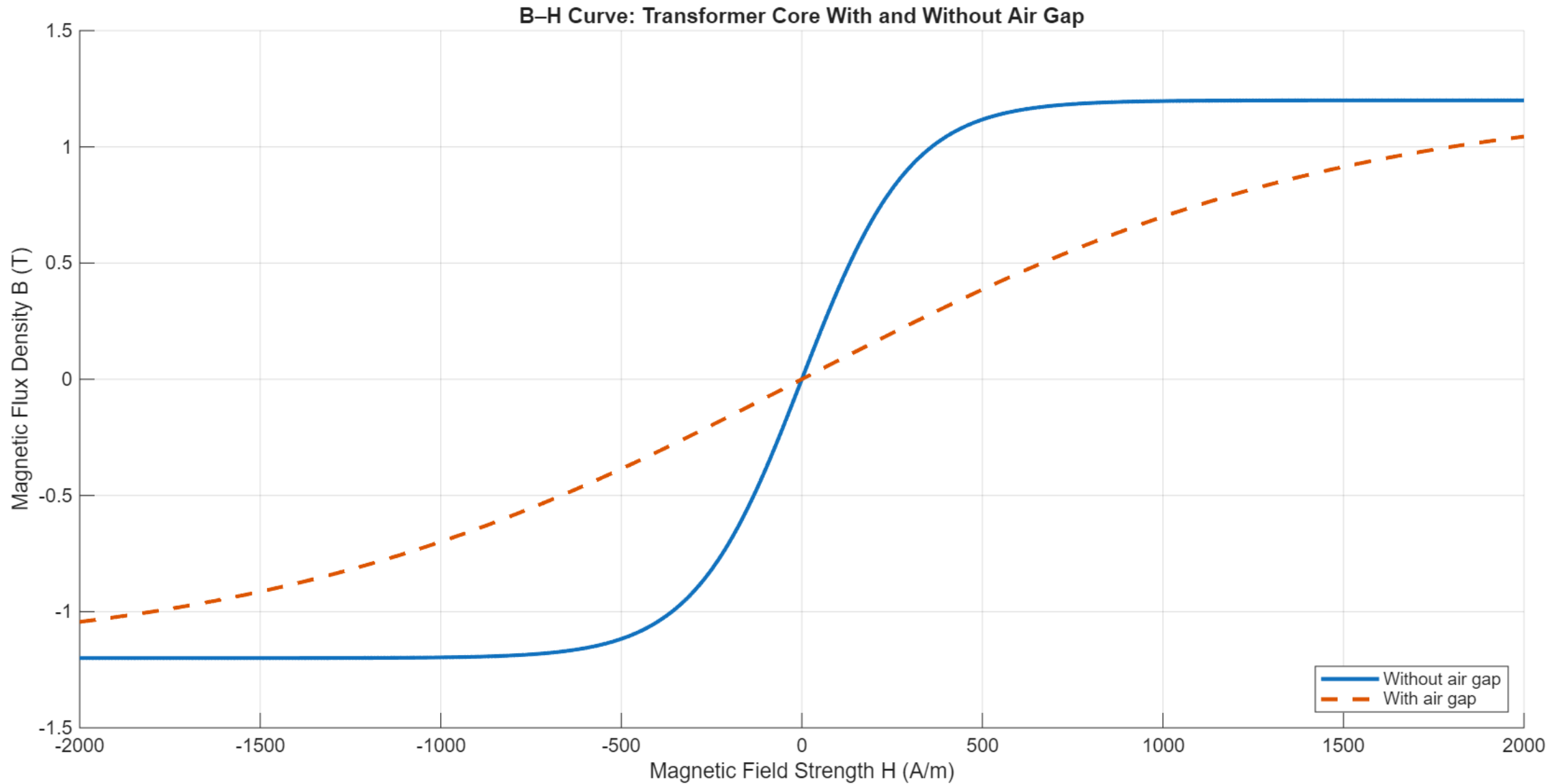
B (magnetic flux density) represents how much magnetic field actually passes through a material — it's the “magnetic response.”

H (magnetic field strength) represents the magnetizing force applied to the material — proportional to the current and number of turns ($N \cdot I/L$).

In a transformer core without an air gap, the B–H curve is steep and saturates quickly, meaning a small increase in current causes a large increase in flux until saturation.

When you add an air gap, the curve becomes flatter and wider: the core can store more magnetic energy and resist saturation, but the effective permeability drops.

FLYBACK TRANSFORMERS



FLYBACK TRANSFORMERS

An ungapped core is ideal for large energy transfer but not for energy storage.

It efficiently passes power from primary to secondary windings through magnetic coupling while holding very little magnetic energy itself. Because of its high permeability, it saturates quickly if asked to store energy.

A gapped core, however, is made for energy storage. The air gap reduces effective permeability, flattens the B–H curve, and allows higher magnetizing current before saturation.

That is why flyback converters use gapped cores, while power transformers use ungapped ones.

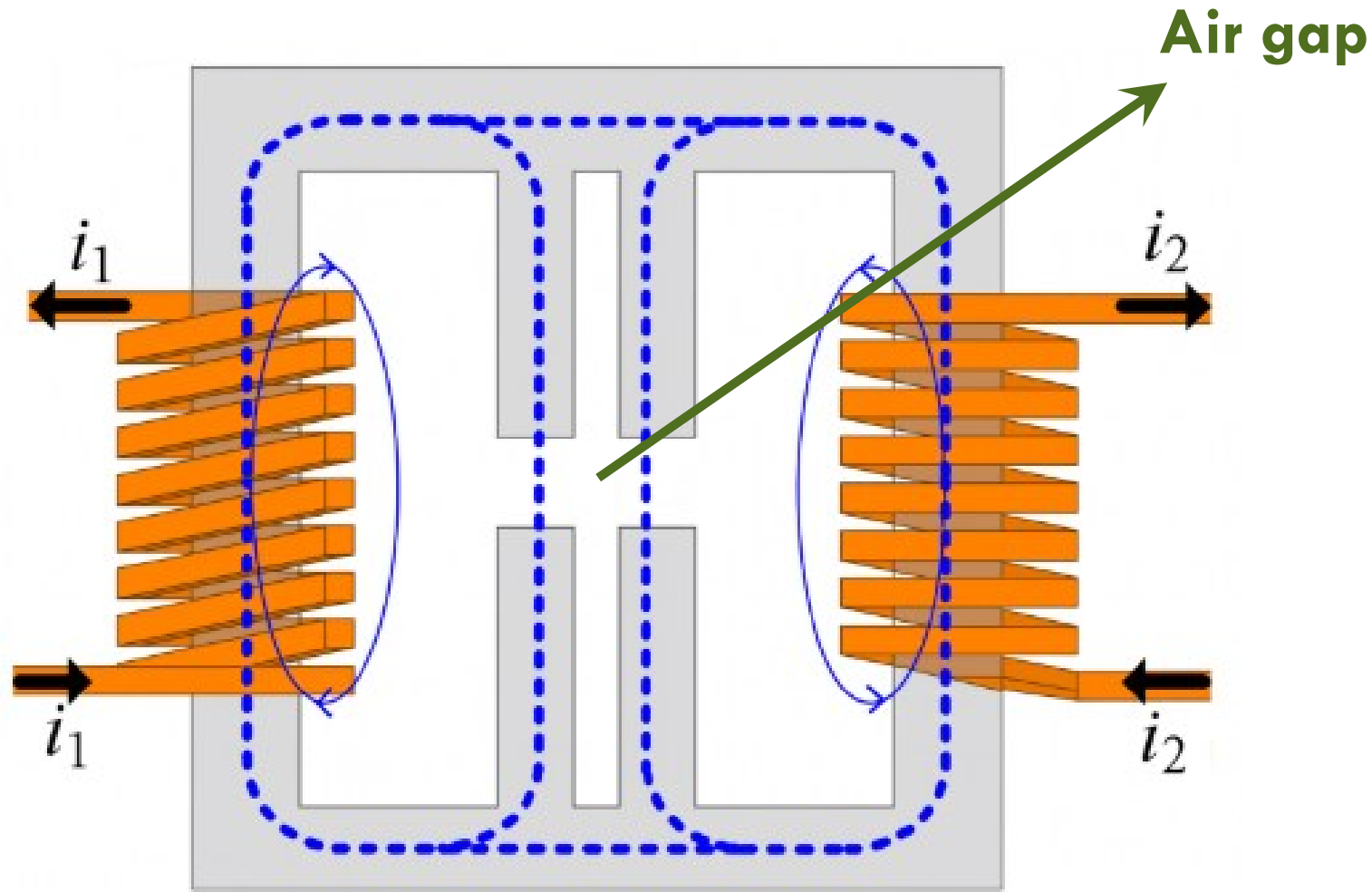
However, the gap also increases magnetic reluctance and weakens the coupling between windings.

As a result, **leakage inductance** becomes higher, meaning part of the magnetic field no longer links both windings completely.

This causes some energy to circulate in stray magnetic paths instead of transferring efficiently, slightly reducing the converter's overall efficiency.

In short, the air gap improves energy storage and linearity but increases leakage and losses, which is why flyback converters accept this trade-off, while power transformers avoid it.

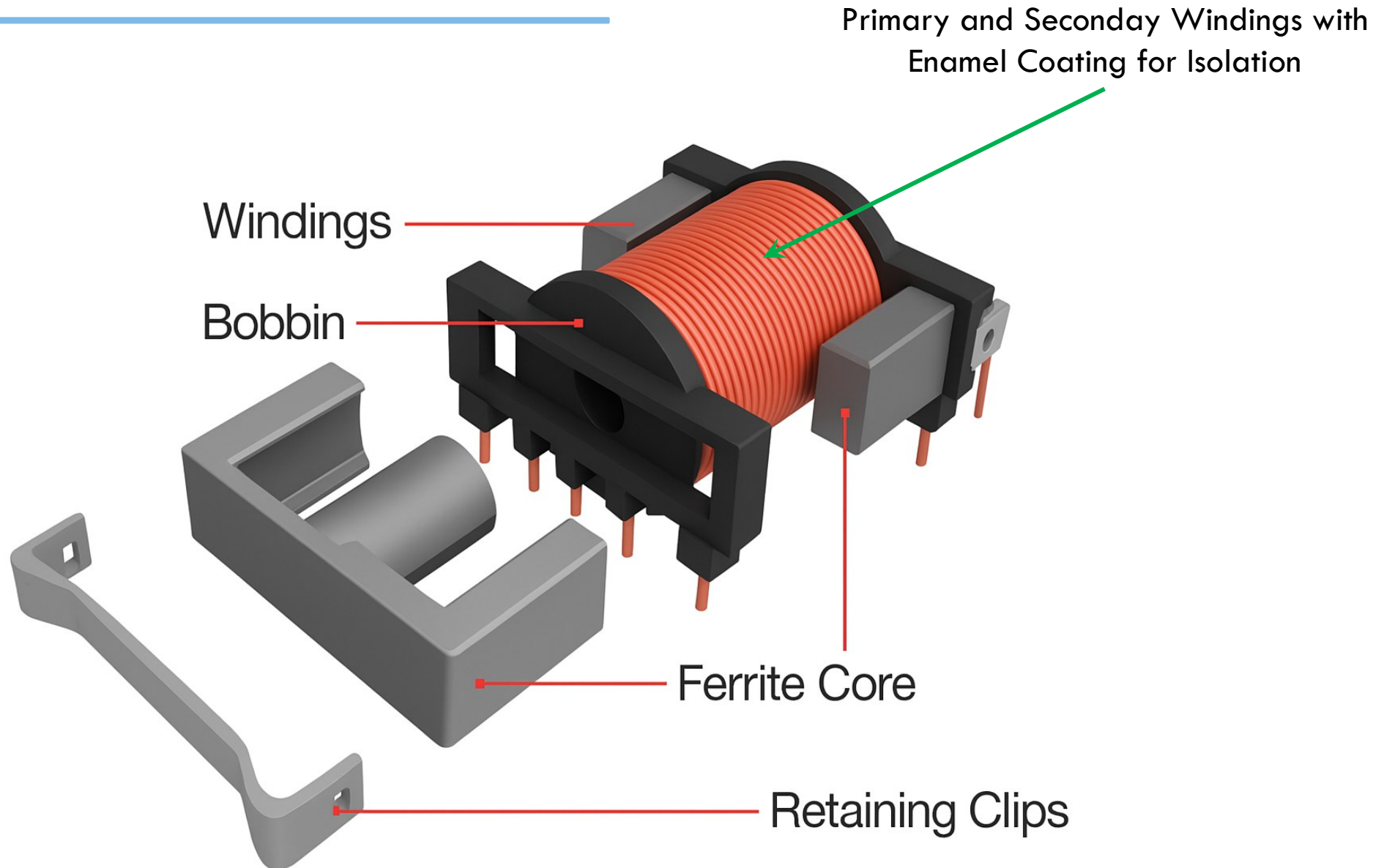
FLYBACK TRANSFORMERS



A Flyback Transformer is actually a pair of coupled inductors. Air gap (in normal transformers, there is no air gap)

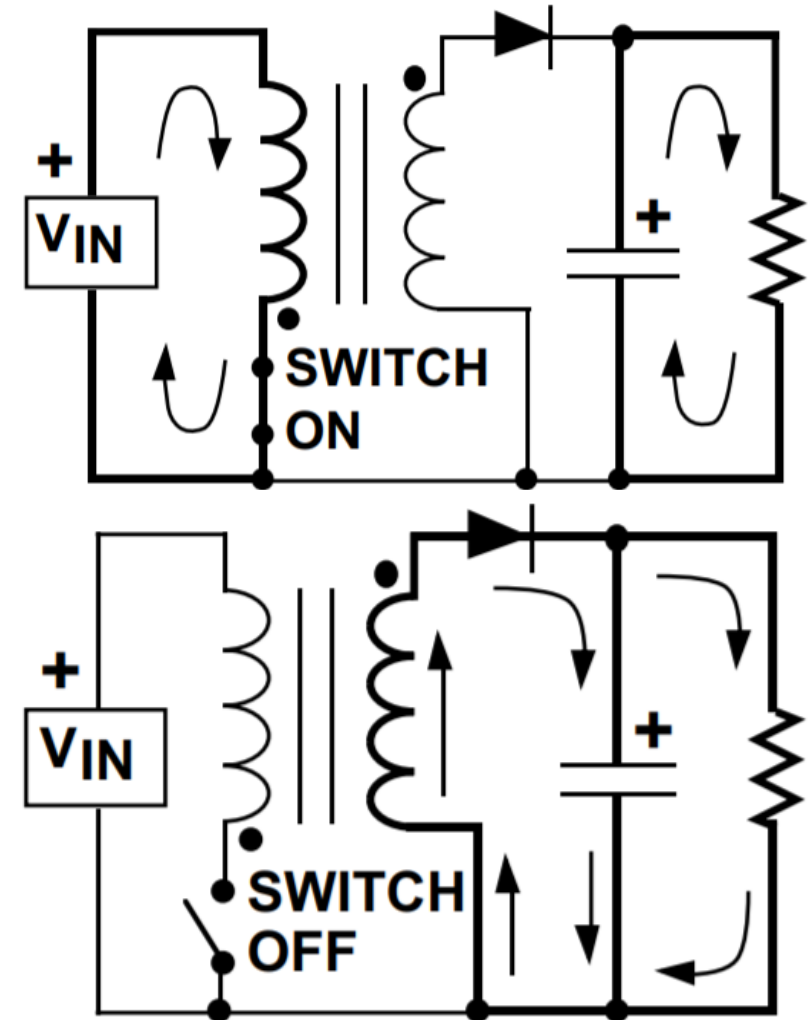
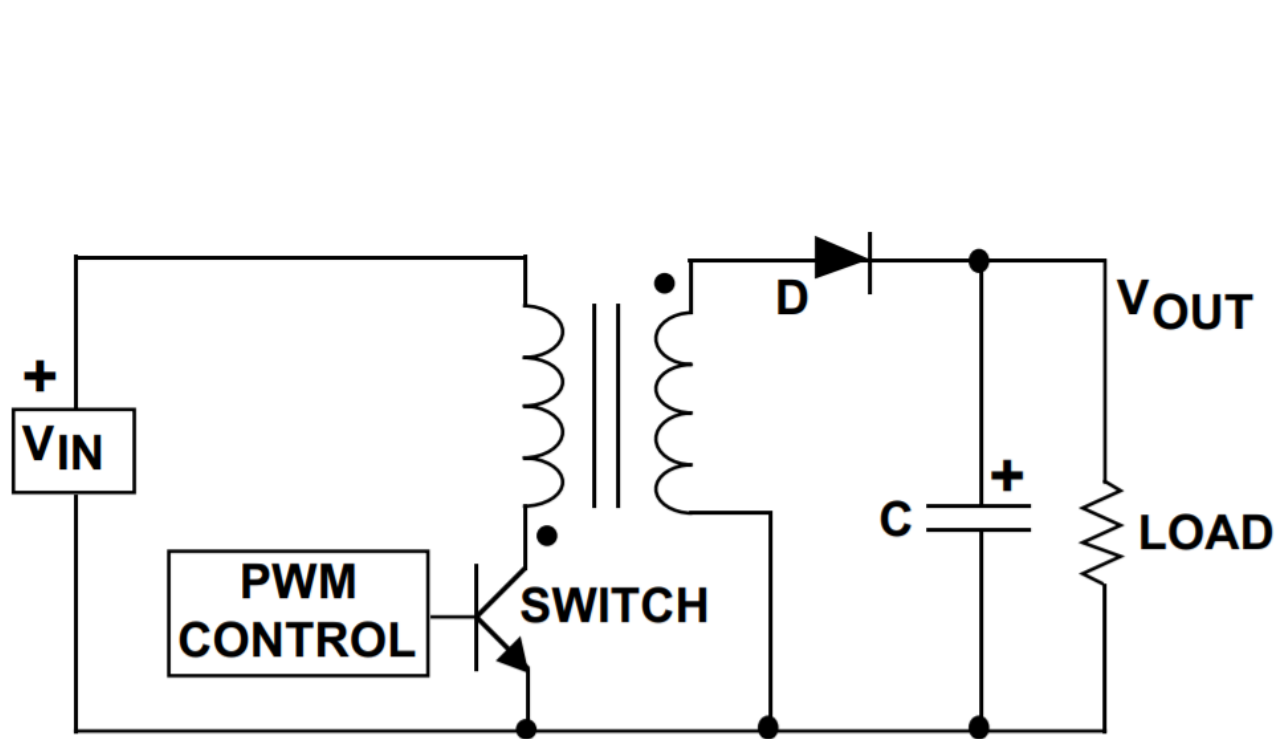
Blue: magnetic flux lines

FLYBACK TRANSFORMERS

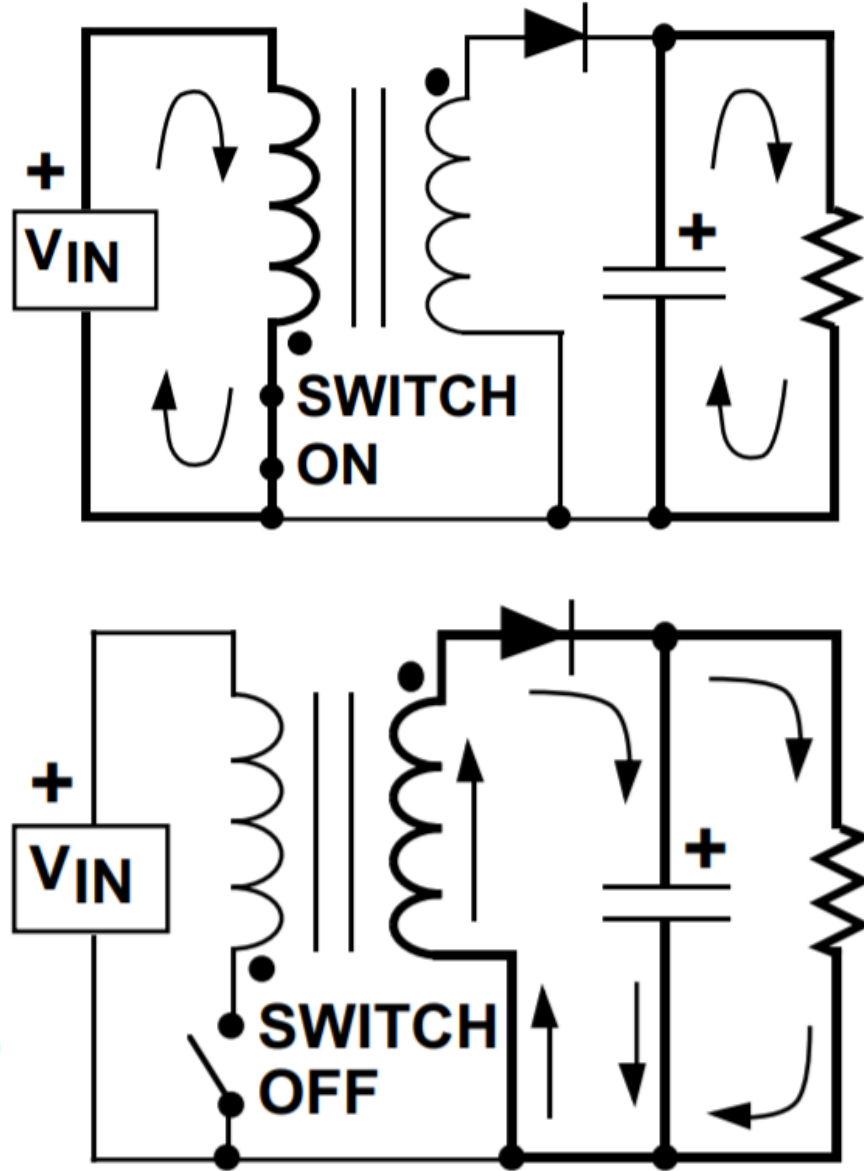


FLYBACK DCDC CONVERTERS

A Flyback regulator converts an input voltage to a higher or lower, positive or negative output voltage level. The energy is transferred to the output when the FET is not conducting. Flyback converters have gained popularity in battery powered systems, where a single voltage must be converted into the required system voltages (for example, +5 V, +12 V and -12 V) with high power conversion efficiency.

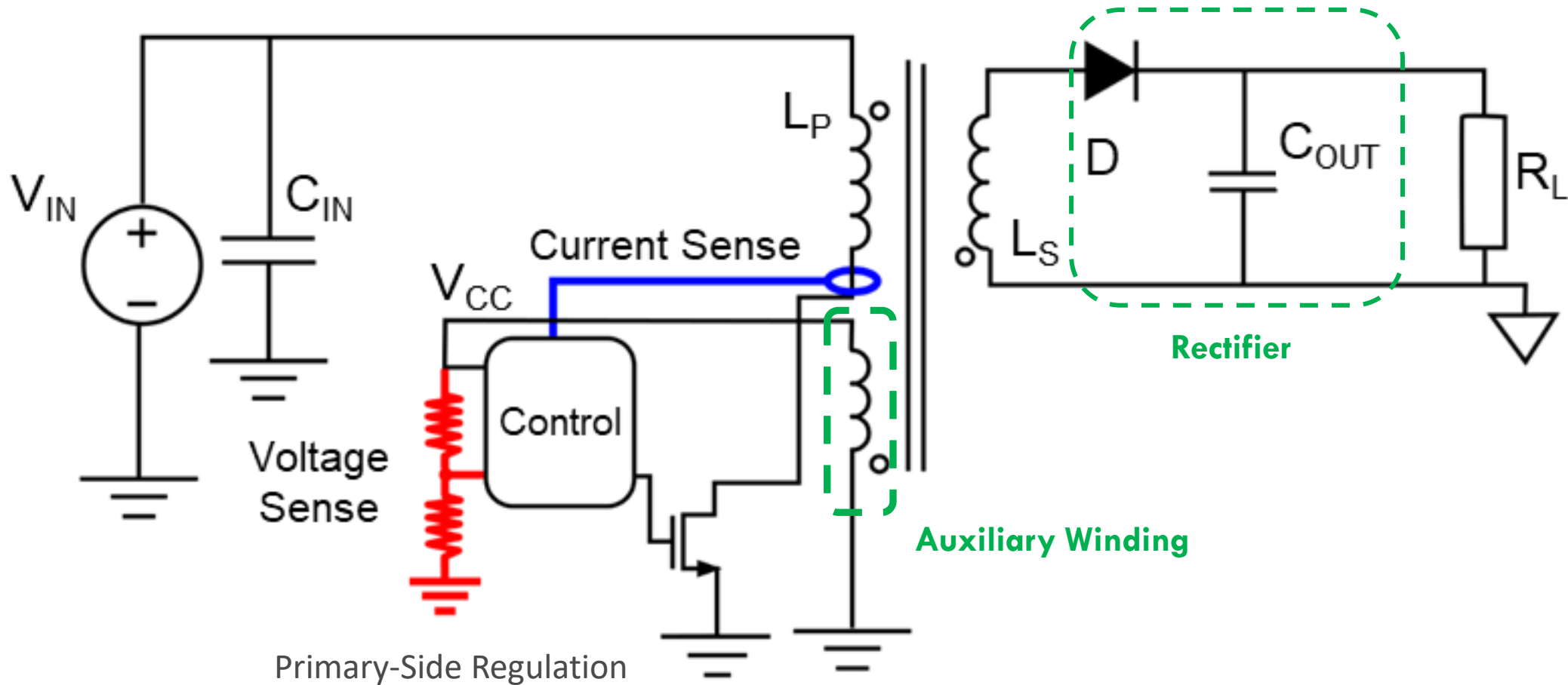


FLYBACK DCDC CONVERTERS



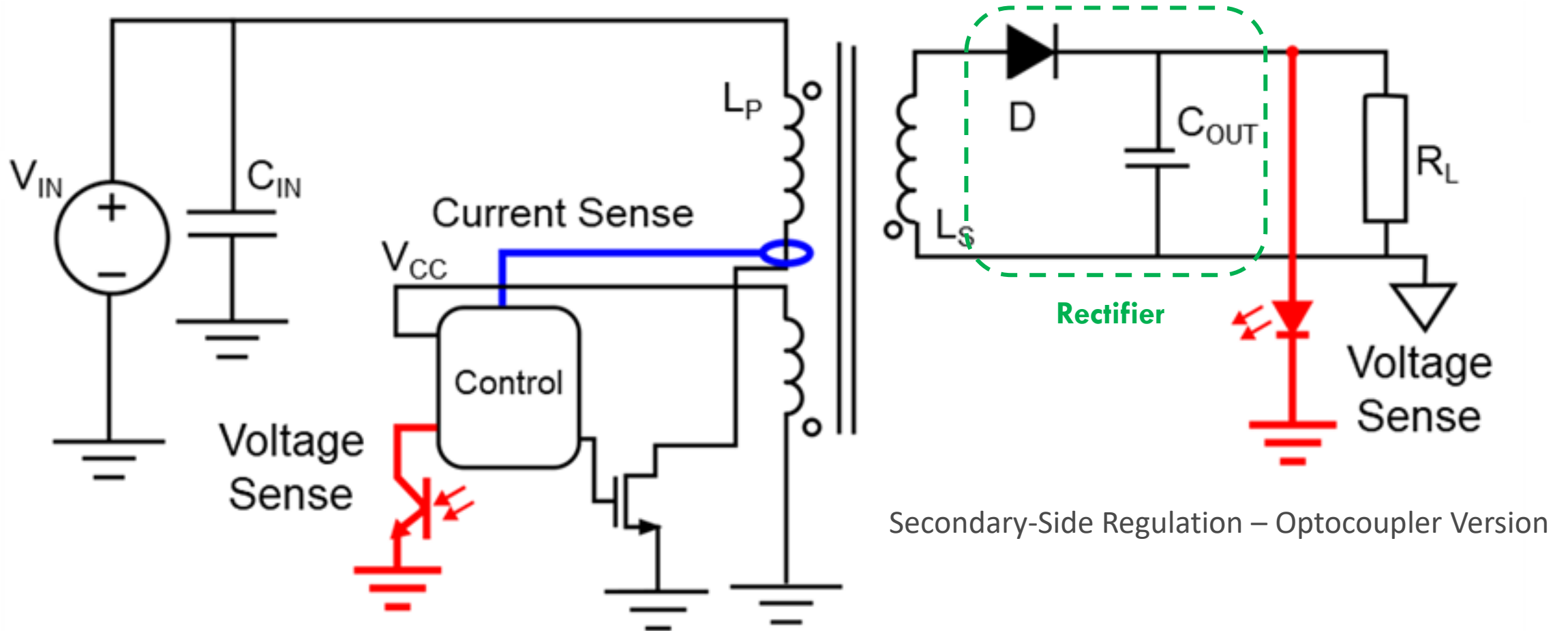
When the switch is on, the input voltage is forced across the transformer primary which causes an increasing flow of current through it. Note that the polarity of the voltage on the primary is dot-negative (more negative at the dotted end), causing a voltage with the same polarity to appear at the transformer secondary (the magnitude of the secondary voltage is set by the transformer secondary-to-primary turns ratio). The dot-negative voltage appearing across the secondary winding turns off the diode, preventing current flow in the secondary winding during the switch on time. During this time, the load current must be supplied by the output capacitor alone. When the switch turns off, the decreasing current flow in the primary causes the voltage at the dot end to swing positive. At the same time, the primary voltage is reflected to the secondary with the same polarity. The dot-positive voltage occurring across the secondary winding turns on the diode, allowing current to flow into both the load and the output capacitor. The output capacitor charge lost to the load during the switch on time is replenished during the switch OFF time. Flyback converters operate in either continuous mode (where the secondary current is always > 0) or discontinuous mode (where the secondary current falls to zero on each cycle).

FLYBACK DCDC CONVERTERS



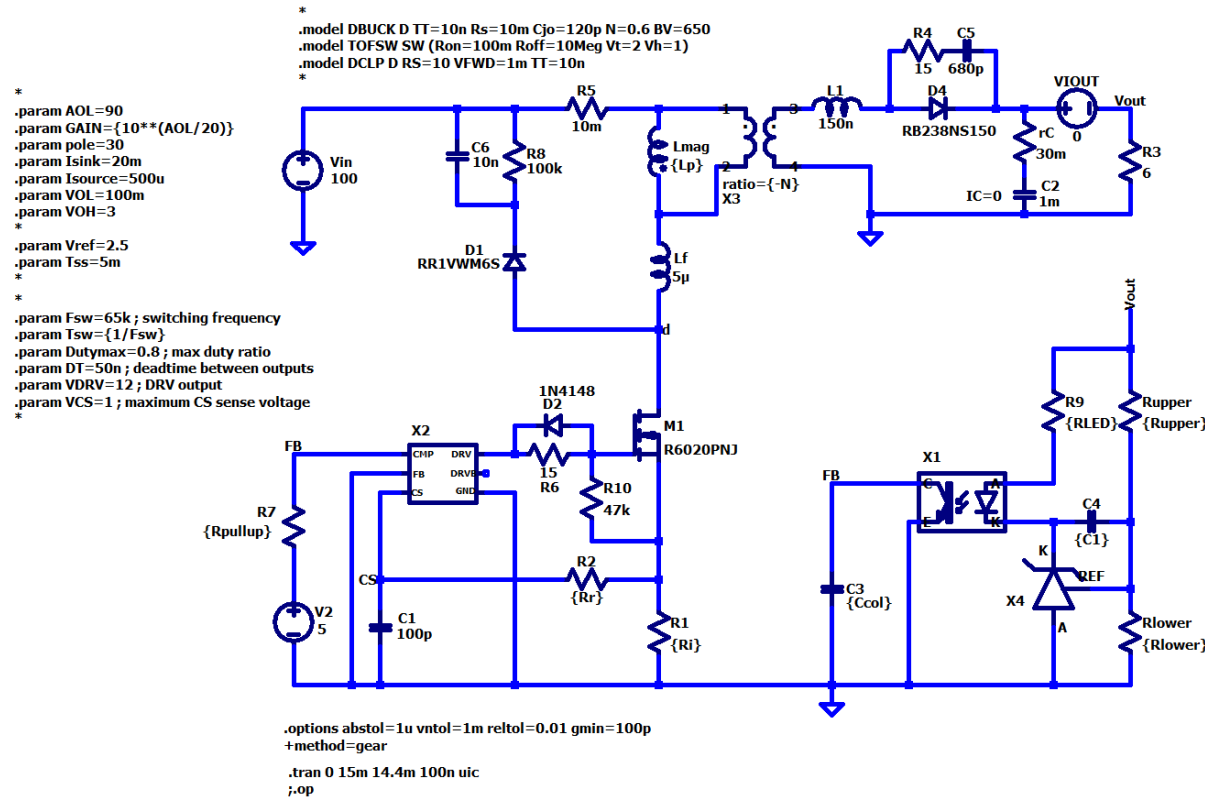
In a flyback converter, the output voltage can be sensed indirectly using an auxiliary winding placed on the transformer core. During the secondary conduction interval, the voltage on the auxiliary winding is proportional to the output voltage according to the turns ratio. By rectifying and filtering this auxiliary voltage, the controller obtains a scaled replica of the output voltage for regulation without using an optocoupler.

FLYBACK DCDC CONVERTERS



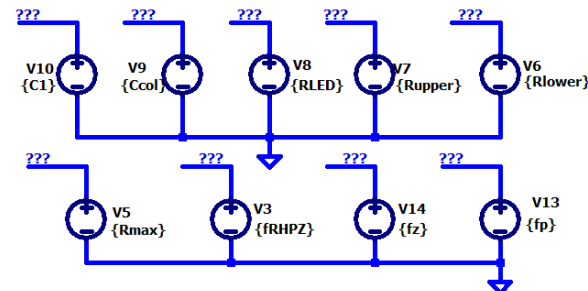
In an optocoupler-based flyback converter, the output voltage is sensed directly on the secondary side using a voltage divider and a reference device such as a TL431. When the output voltage rises above the set point, the TL431 drives current through the optocoupler LED. The optocoupler transistor on the primary side then adjusts the PWM duty cycle, providing isolated feedback regulation.

FLYBACK DCDC CONVERTERS



* Run an operating point to obtain the dc labels (.op)
 * These dc labels are the values of the components:
 pV --> pF
 nV --> nF
 kV --> kOhms
 *

If Ccol is negative, the opto pole is too low



*
 .param Vin=120
 .param Vout=19
 .param Lp=600u
 .param Ri=250m
 .param N=250m
 *

*
 .param Rload=6
 .param D={Vout/(Vout+N*Vin)} ; duty ratio calculation
 .param mc={0.818/(1-D)} ; recommended compensation value for a Q of 1
 .param Sn={ (Vin/Lp)*Ri }
 .param Sramp={2.5/Tsw} ; 2.5 V over Ts - check your IC specs
 .param Se={ (mc-1)*Sn }
 .param Rr={ (Se/Sramp)*19k+1m }
 .param fRHPZ={ ((1-D)**2*Rload/(D*Lp*N**2))/(2*pi) }
 .param fcMAX={0.2*fRHPZ} ; maximum crossover frequency
 *

*
 .param Gfc=-13 ; magnitude at crossover
 .param PS=-80 ; phase lag at crossover
 *

* Enter Design Goals Information Here *

*
 .param fc=2k ; targeted crossover
 .param PM=60 ; choose phase margin at crossover
 *

* Enter the Values for Vout and Bridge Bias Current *

*
 .param Ibias=250u
 .param Rlower={Vref/Ibias}
 .param Rupper={ (Vout-Vref)/Ibias }
 *

* Optocoupler specifications *

*
 .param Rpullup=20k ; check with the selected control chip
 .param Fopto=6k
 .param Copto={1/(2*pi*Fopto*Rpullup)}
 .param CTR=0.33
 .param Rd=100 ; this is the LED dynamic resistance, depends on FB current
 *

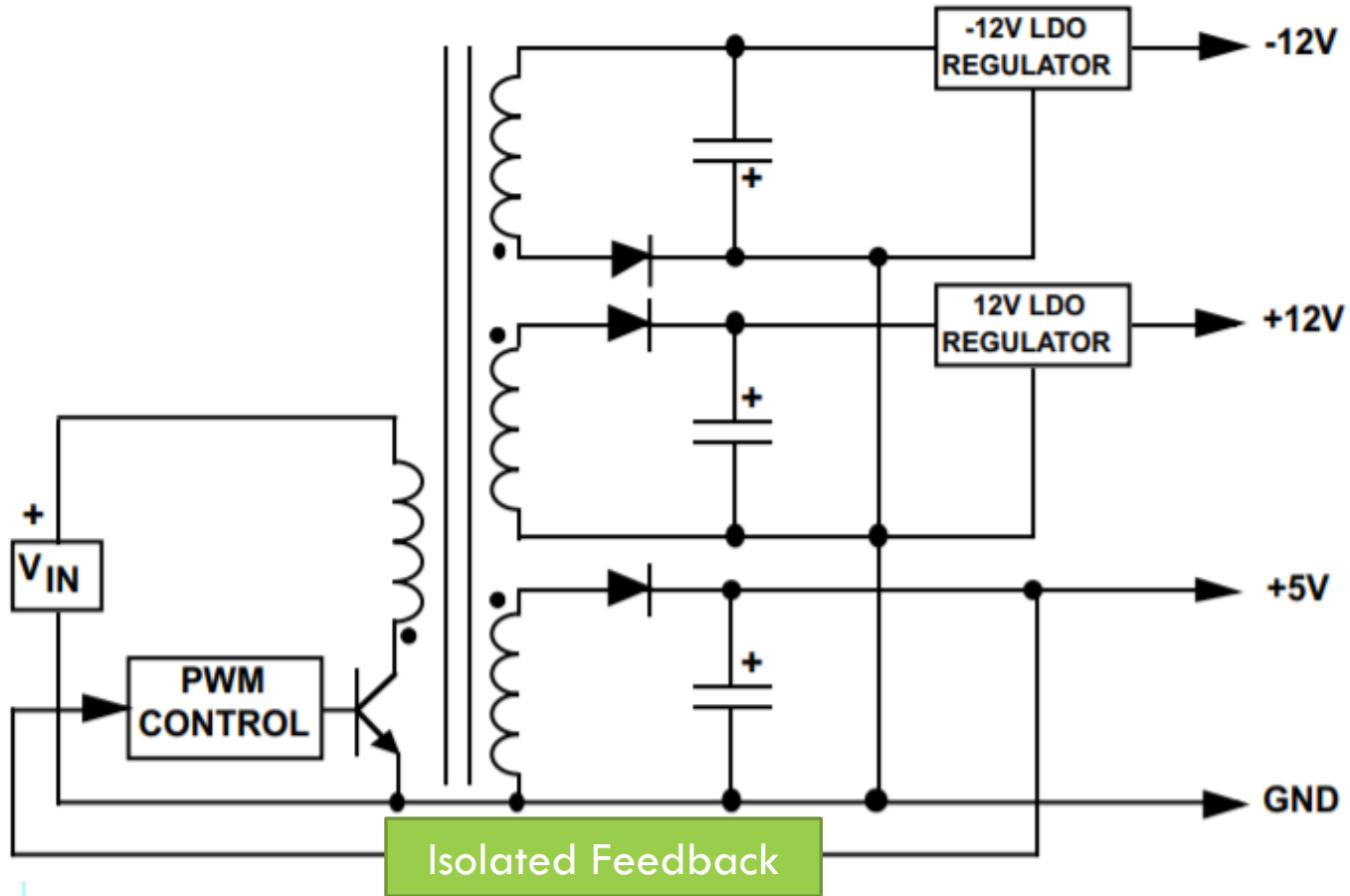
*
 .param VL=0.2
 .param VCESat=0.3
 .param Vdd=5
 .param Vf=1
 .param A={Vout-Vf-VL}
 .param B={Vdd-VCESat}
 .param Rmax={ (A/B)*Rpullup*CTR } ; check RLED is below Rmax
 *

* Do not edit the below lines *

*
 .param boost={PM-PS-90}
 .param fp={ (tan(boost*pi/180)+sqrt((tan(boost*pi/180)**2+1)))*fc }
 .param fz={fc**2/fp}
 .param G={10** (-Gfc/20)}
 .param RLED={CTR*Rpullup/G}
 .param C1={1/(2*pi*fz*Rupper)}
 .param C2={1/(2*pi*fp*Rpullup)}
 .param Ccol={C2-Copto}
 *

Secondary-Side Regulation – Optocoupler Version

FLYBACK DCDC CONVERTERS



Another big advantage of a flyback is the capability of providing multiple outputs. In such applications, one of the outputs (usually the highest current) is selected to provide PWM feedback to the control loop, which means this output is directly regulated.

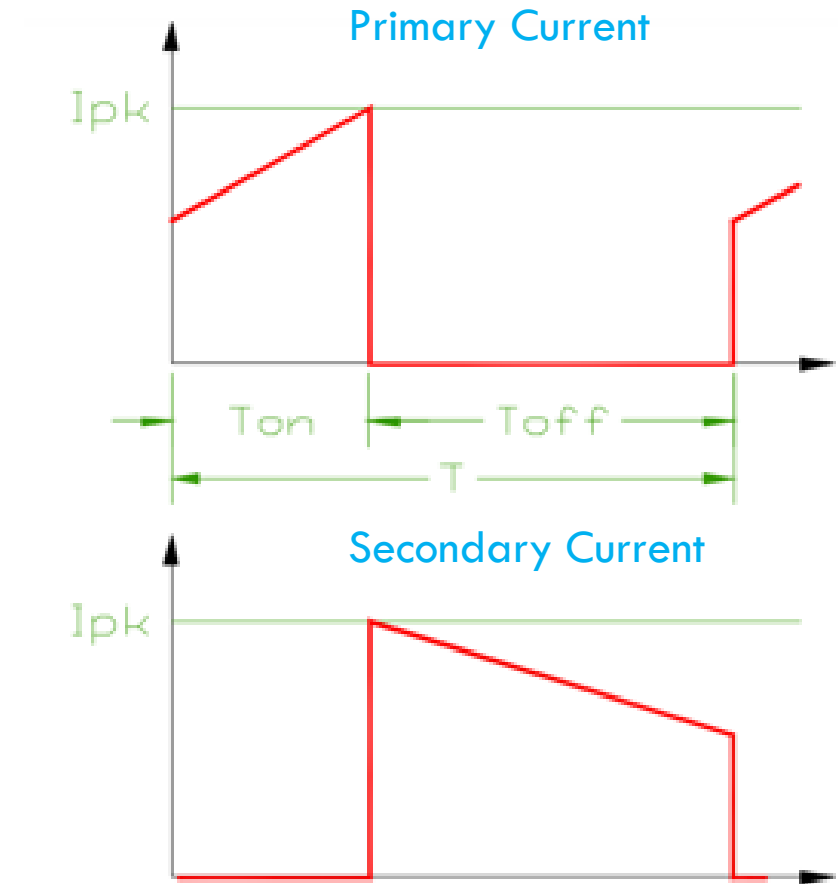
The other secondary winding(s) are indirectly regulated, as their pulse widths will follow the regulated winding. The load regulation on the unregulated secondaries is not great (typically 5% to 10%) but is adequate for many applications.

Typical Multiple-Output Flyback with LDOs

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

CCM: Continuous Conduction Mode

- The primary current never drops to zero during all operating conditions
- Primary inductance value is relatively bigger
- Two states per switching cycle:
 - The ON state: Transistor is ON and Diode is OFF
 - The OFF state: Transistor is OFF and Diode is ON
- Lower ripple currents
- Improved EMC properties



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

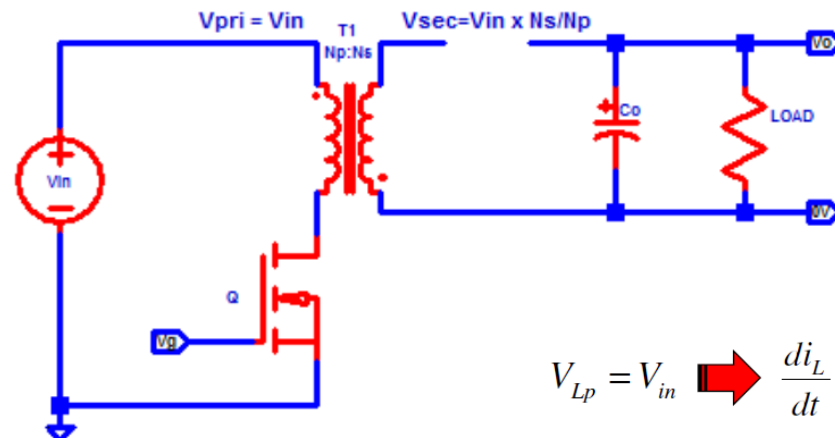
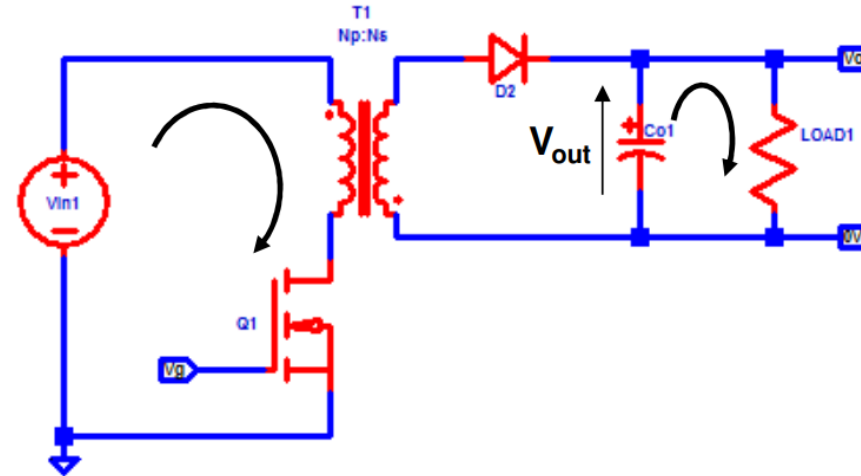
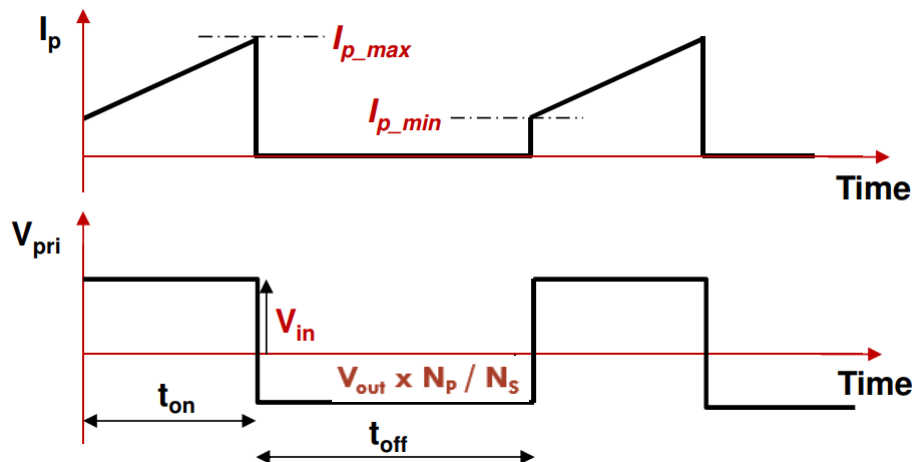
CCM: Continuous Conduction Mode

Switch closed (Transistor ON):

- Primary winding current rises from I_{p_min} to I_{p_max} in $t_{on} = DT$

$$\Delta I_p = \frac{V_{in} \times DT}{L_p}$$

$$I_{p_max} = \frac{V_{out} \times I_{out}}{\eta \times D \times V_{in}} + \frac{\Delta I_p}{2}$$



$$V_{Lp} = V_{in} \Rightarrow \frac{di_L}{dt} = \frac{V_{in}}{L_p}$$

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

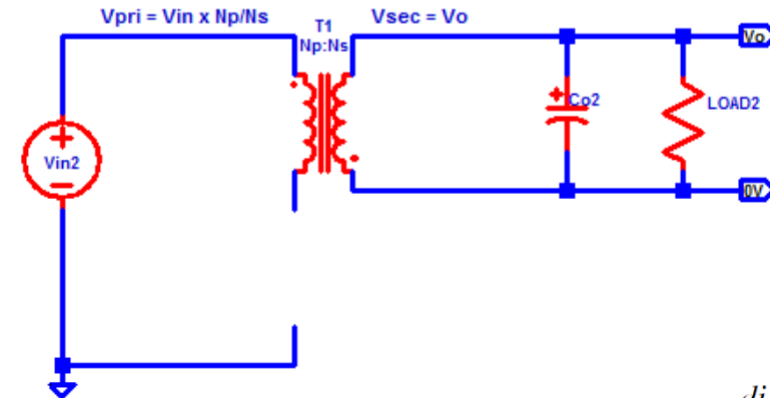
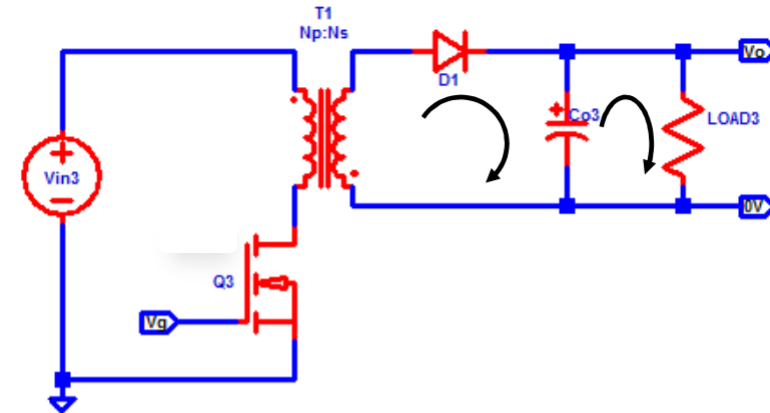
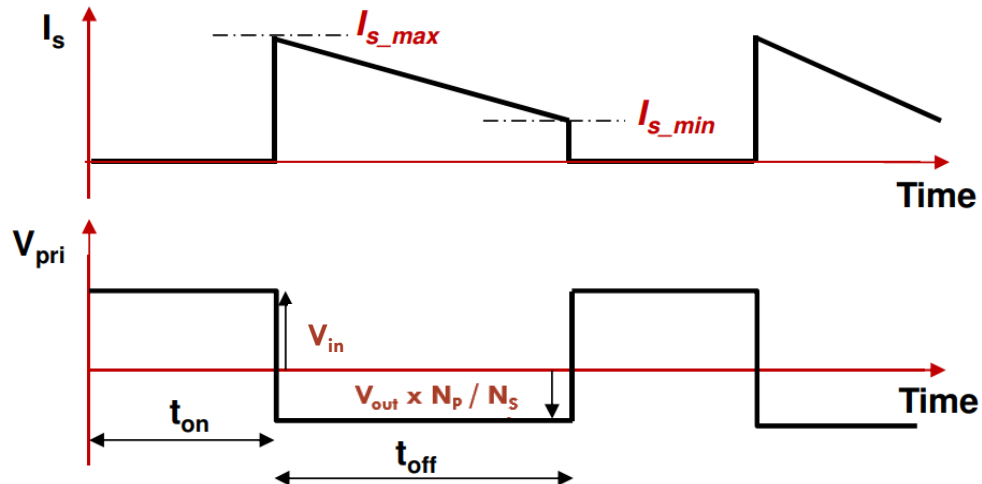
CCM: Continuous Conduction Mode

Switch opened (Transistor OFF):

- Secondary winding current falls from I_{s_max} to I_{s_min} in $t_{off} = (1-D)T$.

$$\Delta I_s = \Delta I_p \times \frac{N_p}{N_s} \quad I_{s_max} = I_{p_max} \times \frac{N_p}{N_s}$$

$$\frac{N_p}{N_s} = \sqrt{\frac{L_p}{L_s}}$$

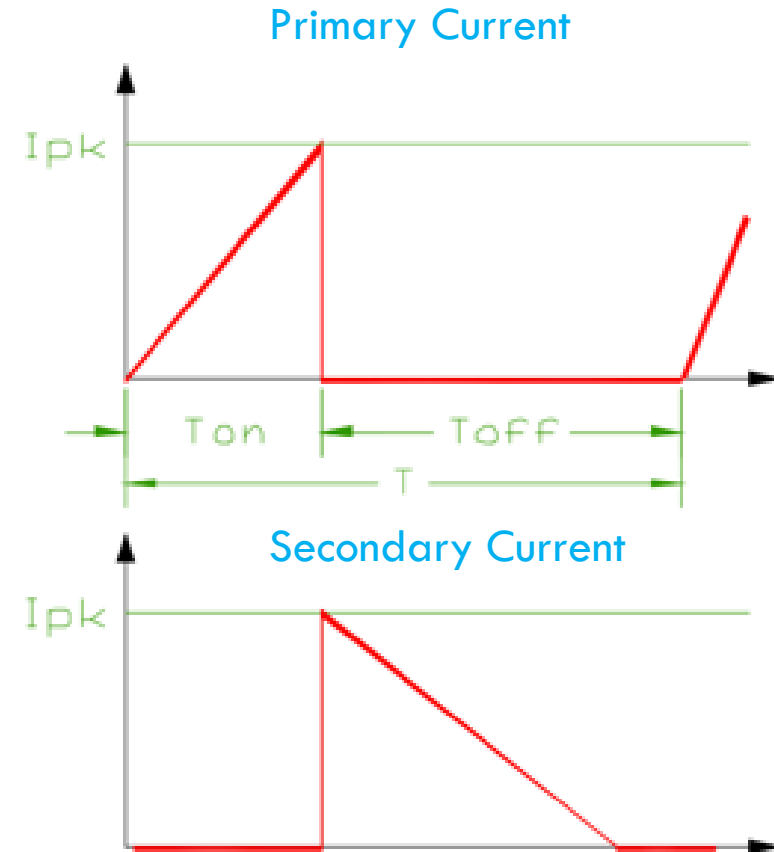


$$V_{Ls} = -V_{out} \Rightarrow \frac{di_{Ls}}{dt} = \frac{-V_{out}}{L_s}$$

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

DCM: Discontinuous Conduction Mode

- The primary current drops to zero.
- Primary Inductance value is relatively smaller.
Capacitance may need to be larger.
- Three states per switching cycle:
 - The ON state: Transistor is ON and Diode is OFF
 - The OFF state: Transistor is OFF and Diode is ON
 - The IDLE state: Both Transistor and Diode are OFF
- MOSFET turn on at zero current



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

DCM: Discontinuous Conduction Mode

- Switch closed (Transistor ON):

- Primary winding current rises from zero to I_{p_max} in $t_{on} = DT$

$$I_{p_max} = \Delta I_p = \frac{V_{in} \times DT}{L_p}$$

$$D = \frac{\sqrt{2 \times P_{in} \times L_p \times f}}{V_{in}}$$

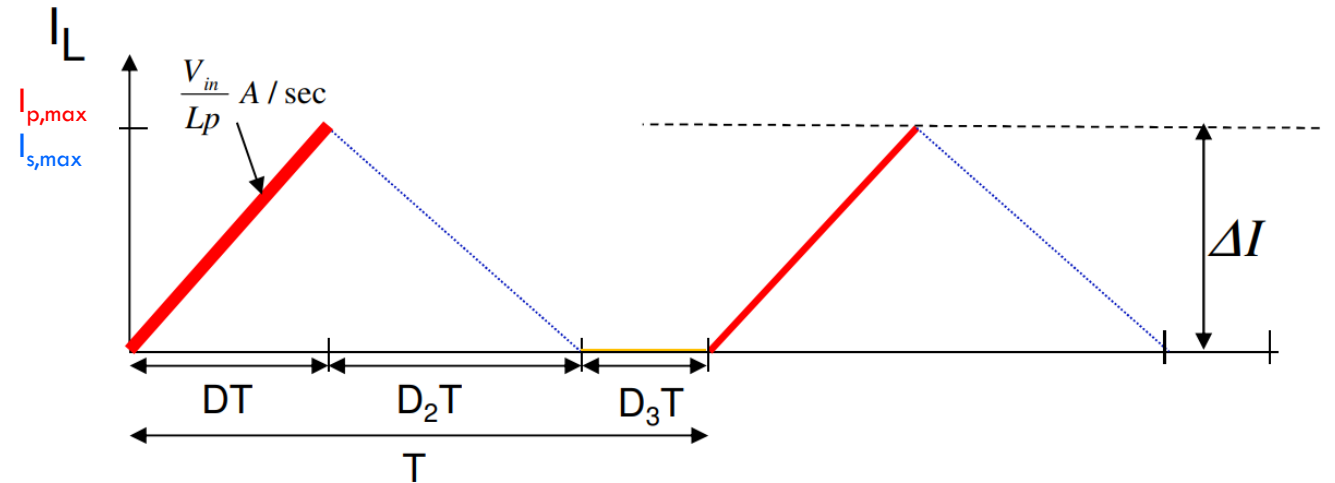
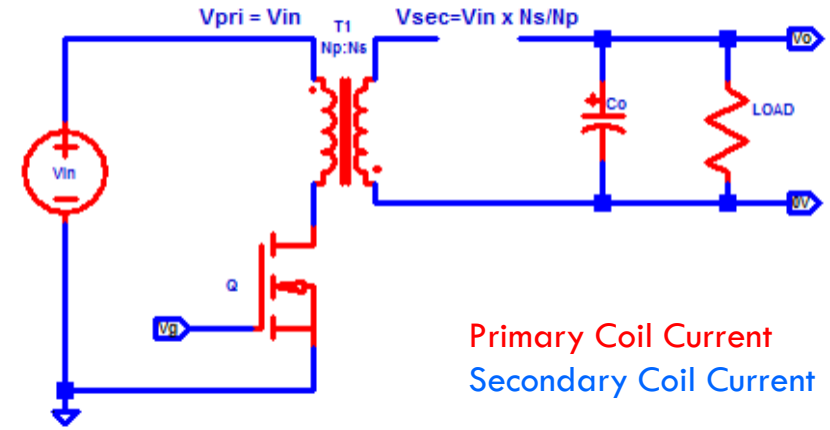
from

$$E = \frac{1}{2} L_p I_{p,max}^2$$

$$E = \frac{1}{2} L_p \left(\frac{V_{in} D}{L_p f} \right)^2$$

where

$$P_{in} = E \cdot f$$



Magnetization Current I_L of the Flyback Transformer
(Primary and Secondary Current Combined)

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

DCM: Discontinuous Conduction Mode

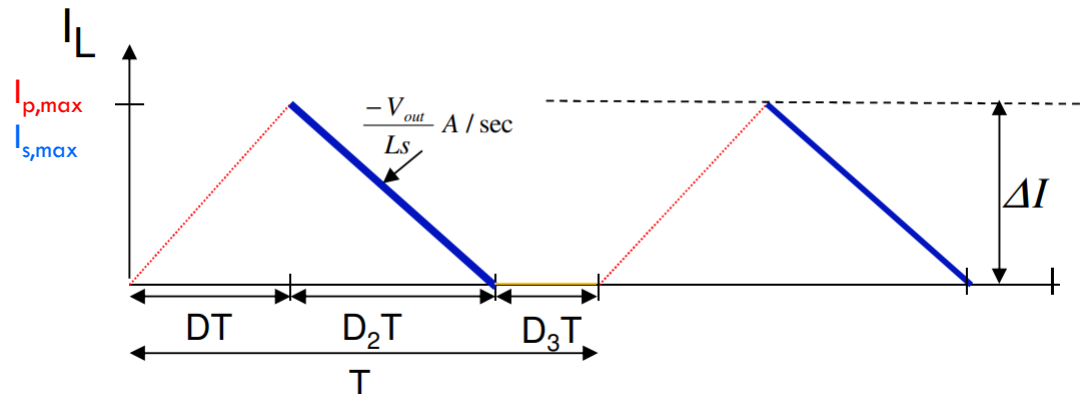
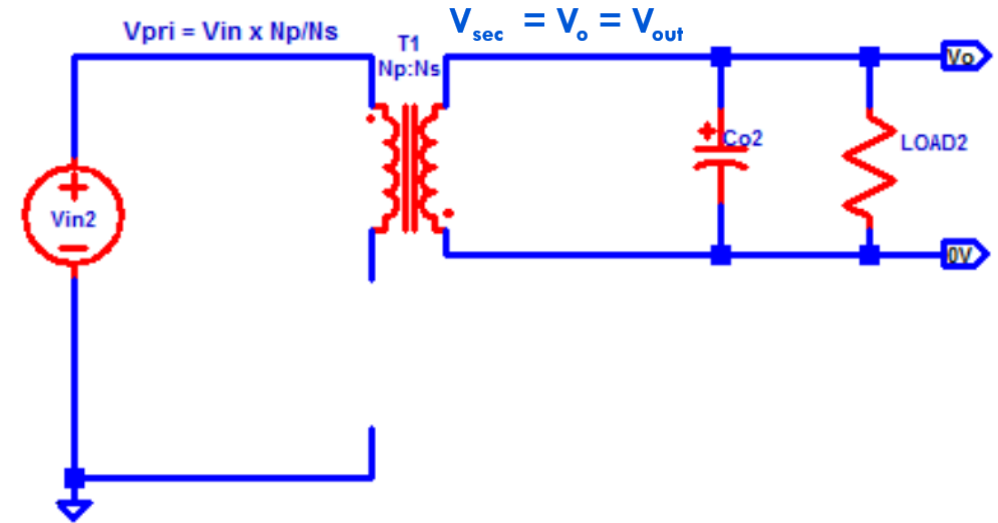
- Switch opened (Transistor OFF):

- Secondary winding current falls from I_{s_max} to zero in $t_{off} = D_2 T$.

$$I_{s_max} = I_{p_max} \times \frac{N_p}{N_s}$$

$$D_2 = \frac{\sqrt{2 \times P_{out} \times N^2 \times L_p \times f}}{V_{out}}$$

$$N = \frac{N_s}{N_p}$$

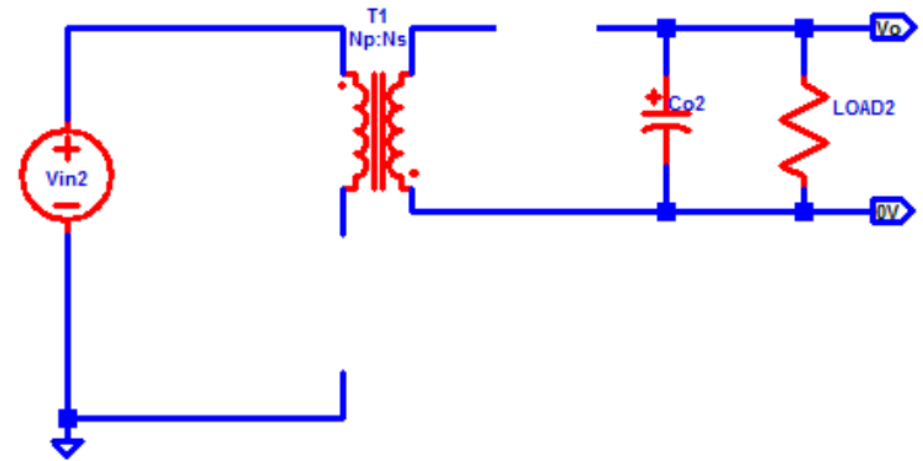
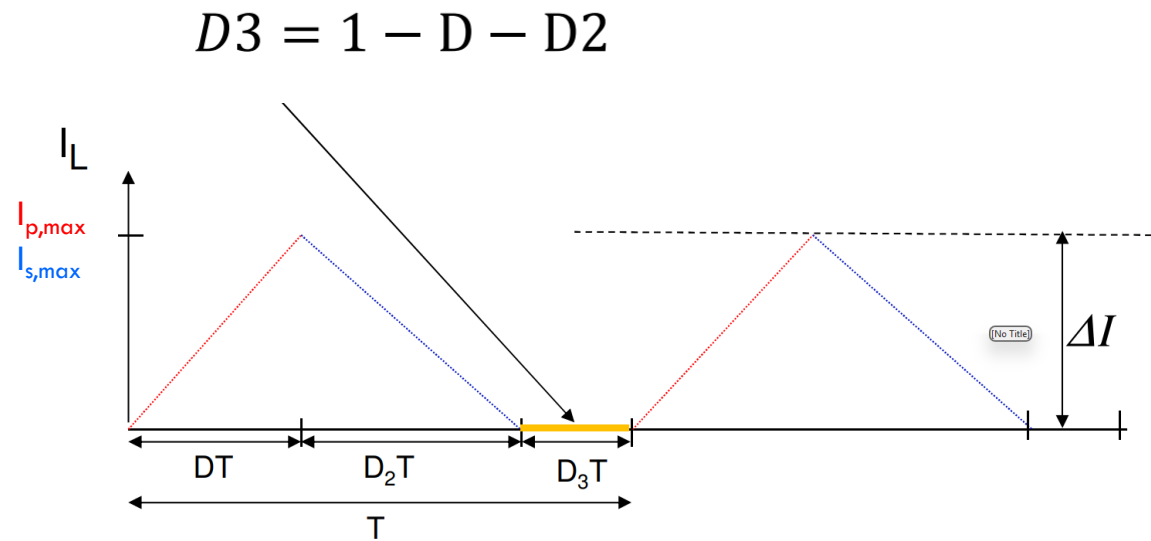


FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

DCM: Discontinuous Conduction Mode

Idle State (Transistor & Diode OFF):

- Only Capacitor delivers energy to load during D_3T



FLYBACK DCDC CONVERTER – EXAMPLE

Flyback DCDC Converter Design Example - DCM

V_{in} : Nominal range 200 V to 800 V

V_{out} : 12 V

P_{out} : 62 W

For DCM: Selecting a switching frequency of 150 kHz and duty cycle of 0.5, and assuming a 95% efficiency at input voltage of 200 V, the maximum primary inductance is found using

$$P_{in} = \frac{P_o}{\eta} = \frac{D^2 V_{in}^2}{2 L_{pri} F_{SW}}$$

$$I_{pp1} = \frac{V_{in} \cdot D}{L_{pri} \cdot F_{SW}}$$

Where P_o is the total output power, η is the assumed efficiency, I_{pp1} is primary peak-peak current, L_{pri} is the flyback transformer primary winding inductance, and F_{SW} is the switching frequency.

FLYBACK DCDC CONVERTER – EXAMPLE

Flyback DCDC Converter Design Example - DCM

The maximum L_{pri} can be found from the previous equation:

$$L_{pri} = \frac{\eta D^2 V_{in}^2}{2 F_{SW} P_{out}}$$

$$L_{pri} = \frac{0.95 \cdot (0.5)^2 \cdot (200)^2}{2 \cdot 62 \cdot 150 \text{ k}} = 511 \mu\text{H}$$

Ignore the voltage losses in the output rectification diode and resistive elements, input-output voltage regulation can be represented by

$$(1 - D) N_{ps} V_{out} = D V_{in_nom_min}$$

$$N_{ps} = \frac{0.5 \times 200}{(1 - 0.5) \times 12} = 16.7$$

To avoid fractional turns of the auxiliary winding, an integer number 16 is selected in this design. The transformer turns-ratio is:



$$N_{pri}:N_{sec}:N_{aux} = 16:1:1$$

$$L_{sec} = L_{aux} = \frac{511}{16^2} = 2 \mu\text{H}$$

FLYBACK DCDC CONVERTER – EXAMPLE

Flyback DCDC Converter Design Example - DCM

The maximum voltage stress on the power switch happens when it is turned off

$$V_{dsmax} = 800 + 16 \times 12 = 992 \text{ V}$$



FLYBACK DCDC CONVERTER – EXAMPLE

Flyback DCDC Converter Design Example - CCM

A flyback DC–DC converter operates in continuous conduction mode (CCM) with the following specifications:

- Minimum input voltage: $V_{in,min} = 100 \text{ V}$
 - Output power: $P_{out} = 120 \text{ W}$
 - Converter efficiency: $\eta = 0.9$
 - Switching frequency: $f_{sw} = 100 \text{ kHz}$
 - Maximum duty cycle at $V_{in,min}$: $D_{max} = 0.45$
 - Desired primary current ripple: 30% of average primary current
-
- Calculate the average primary current.
 - Calculate the primary current ripple.
 - Determine the required primary inductance L_{pri} .
 - Verify that the converter operates in CCM.



FLYBACK DCDC CONVERTER – EXAMPLE

Flyback DCDC Converter Design Example - CCM

Input power:

$$P_{in} = P_{out} / \eta$$

$$P_{in} = 120 / 0.9$$

$$P_{in} = 133 \text{ W}$$

Average primary current at minimum input voltage:

$$I_{p,avg} = P_{in} / V_{in_min}$$

$$I_{p,avg} = 133 / 100$$

$$I_{p,avg} = 1.33 \text{ A}$$

Ripple current is chosen as 30% of the average current:

$$\Delta I_p = 0.3 \cdot I_{p,avg}$$

$$\Delta I_p = 0.3 \cdot 1.33$$

$$\Delta I_p = 0.40 \text{ A}$$



FLYBACK DCDC CONVERTER – EXAMPLE

Flyback DCDC Converter Design Example - CCM

CCM ripple equation:

$$I_p = (V_{in,min} \cdot D_{max}) / (L_{pri} \cdot f_{sw})$$

Rearranging for Lpri:

$$L_{pri} = (V_{in,min} \cdot D_{max}) / (\Delta I_p \cdot f_{sw})$$

Substitute values:

$$L_{pri} = (100 \cdot 0.45) / (0.40 \cdot 100000)$$

$$L_{pri} = 45 / 40000 \cdot L_{pri} = 1.125 \text{ mH}$$

Minimum primary current:

$$I_{p,min} = I_{p,avg} - \Delta I_p / 2 \quad I_{p,min} = 1.33 - 0.20$$

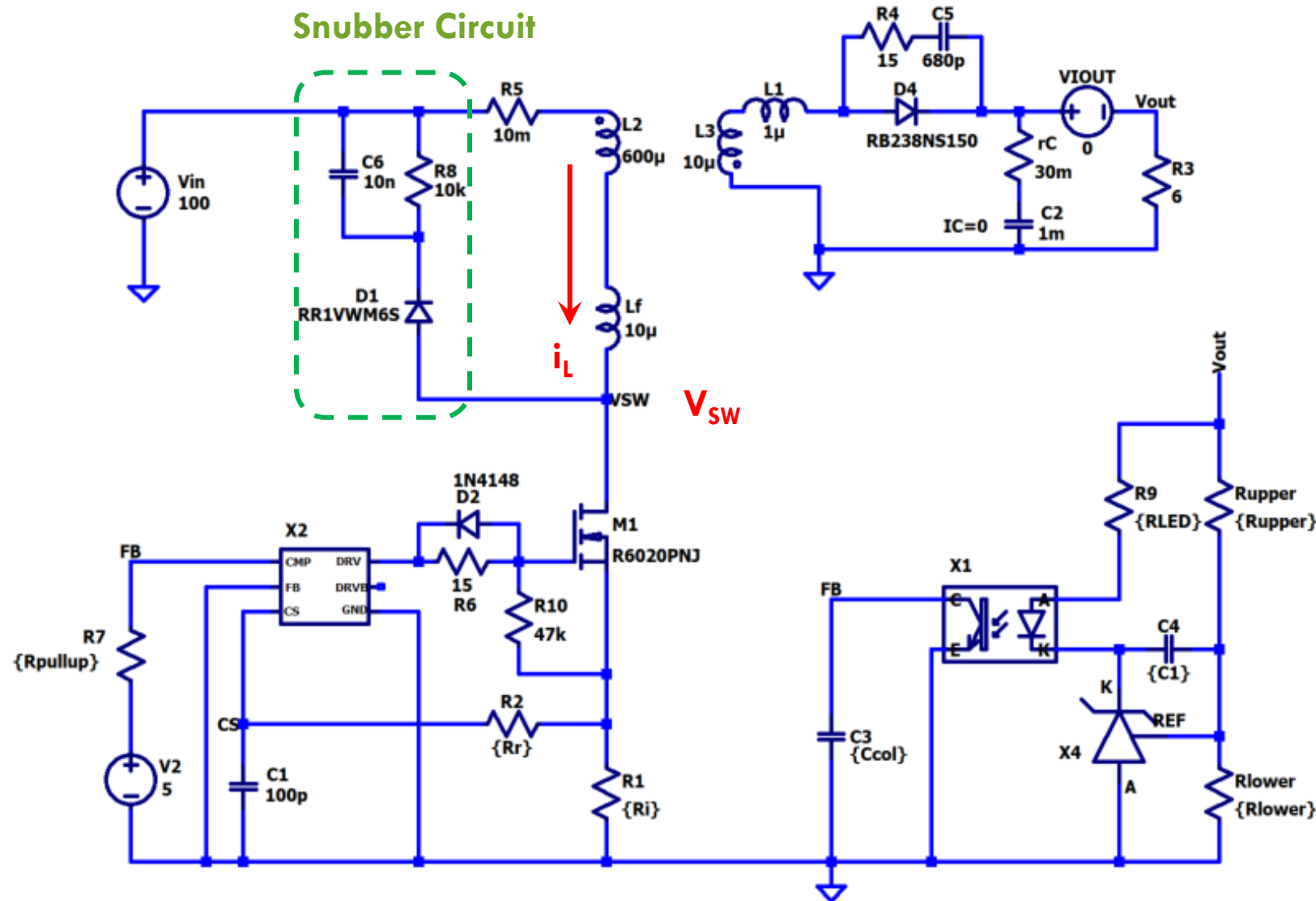
$$I_{p,min} = 1.13 \text{ A}$$

Since $I_{p_min} > 0$, the converter operates in CCM.

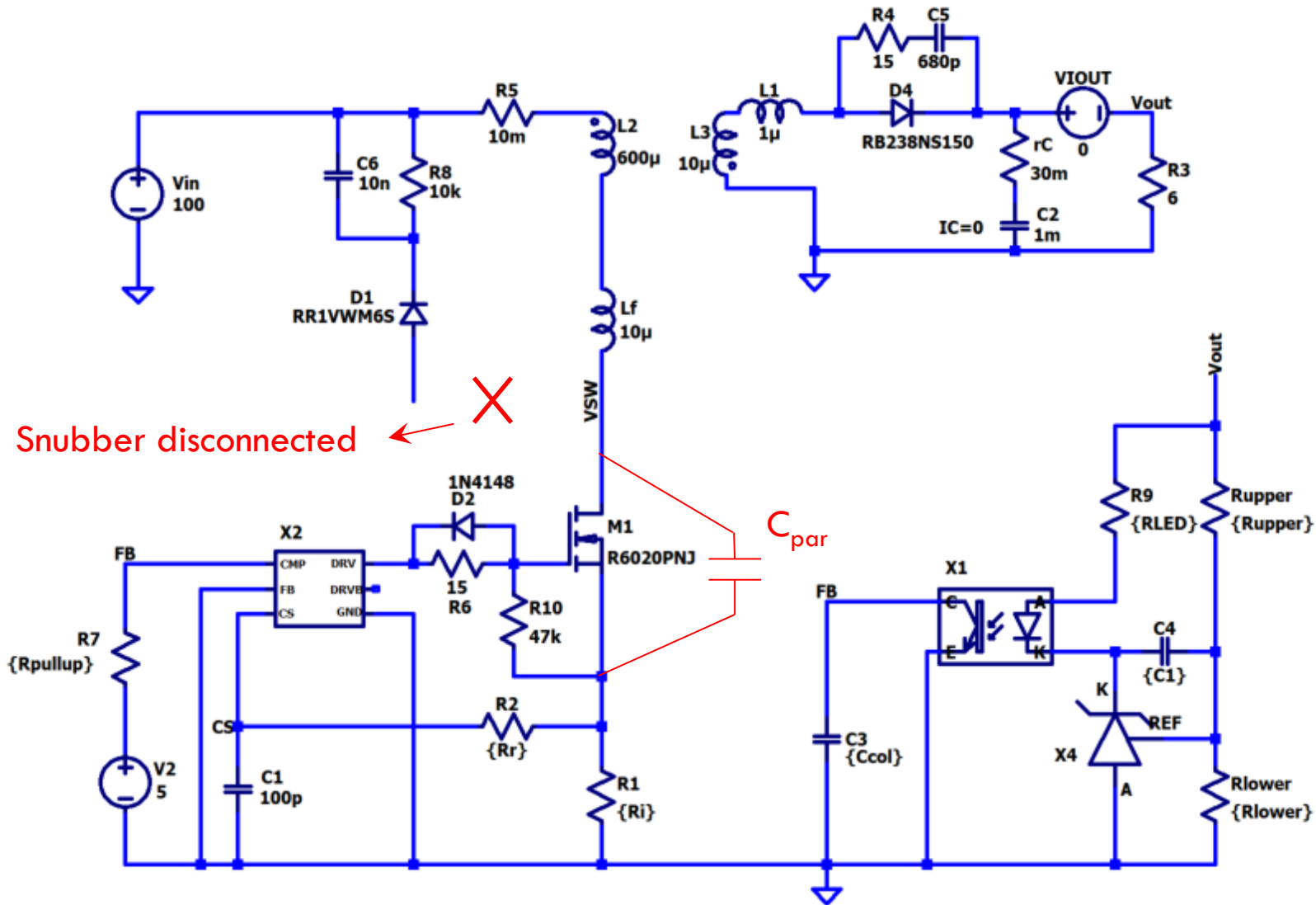
- 1) In reality, the transformer has so-called stray inductances (L_f), that cause oscillations that may damage the transistor M1 (Breakdown Voltage of M1 = 600 V). When M1 turns off, L_{mag} tries to maintain the current according to:

If Q1 turns off, VSW increases abruptly according to:

2) The internal transistor Q1, that drives L1 has also a parasitic capacitance C_{par} between the nodes SW and Source, which might cause additional oscillations (in some cases, C_{par} might suppress the voltage increase due to the stray inductances!).



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE



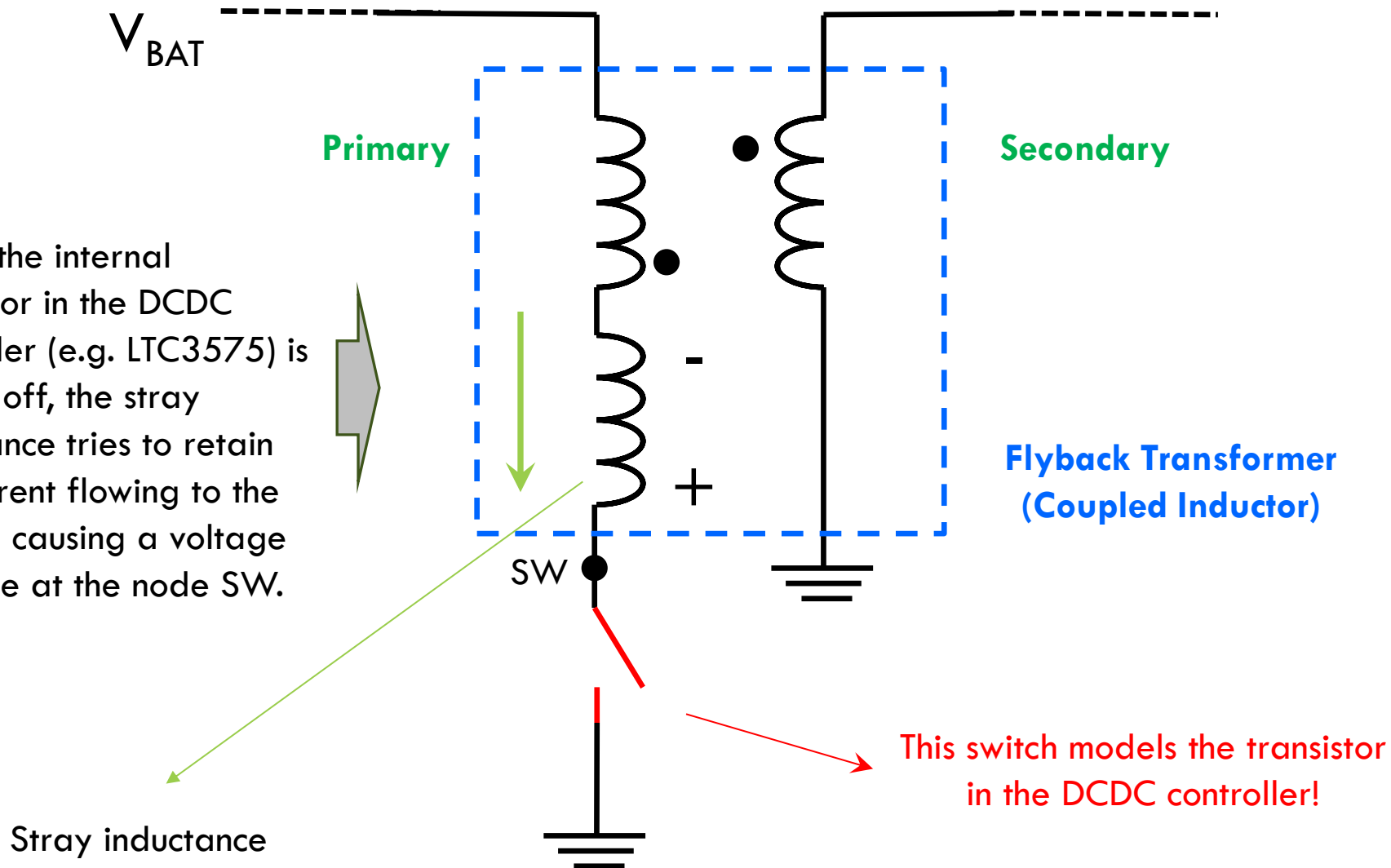
L_f and L_1 are stray inductances of the flyback transformer formed by L_2 and L_3 . C_5 and R_4 form a snubber circuit for the diode D_4 . The internal capacitance of the D_4 can cause an oscillation together with L .

C_{par} is the parasitic capacitance of the transistor M_1 . C_{par} might cause oscillations together the stray inductance L_f .

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Effect of Stray Inductance

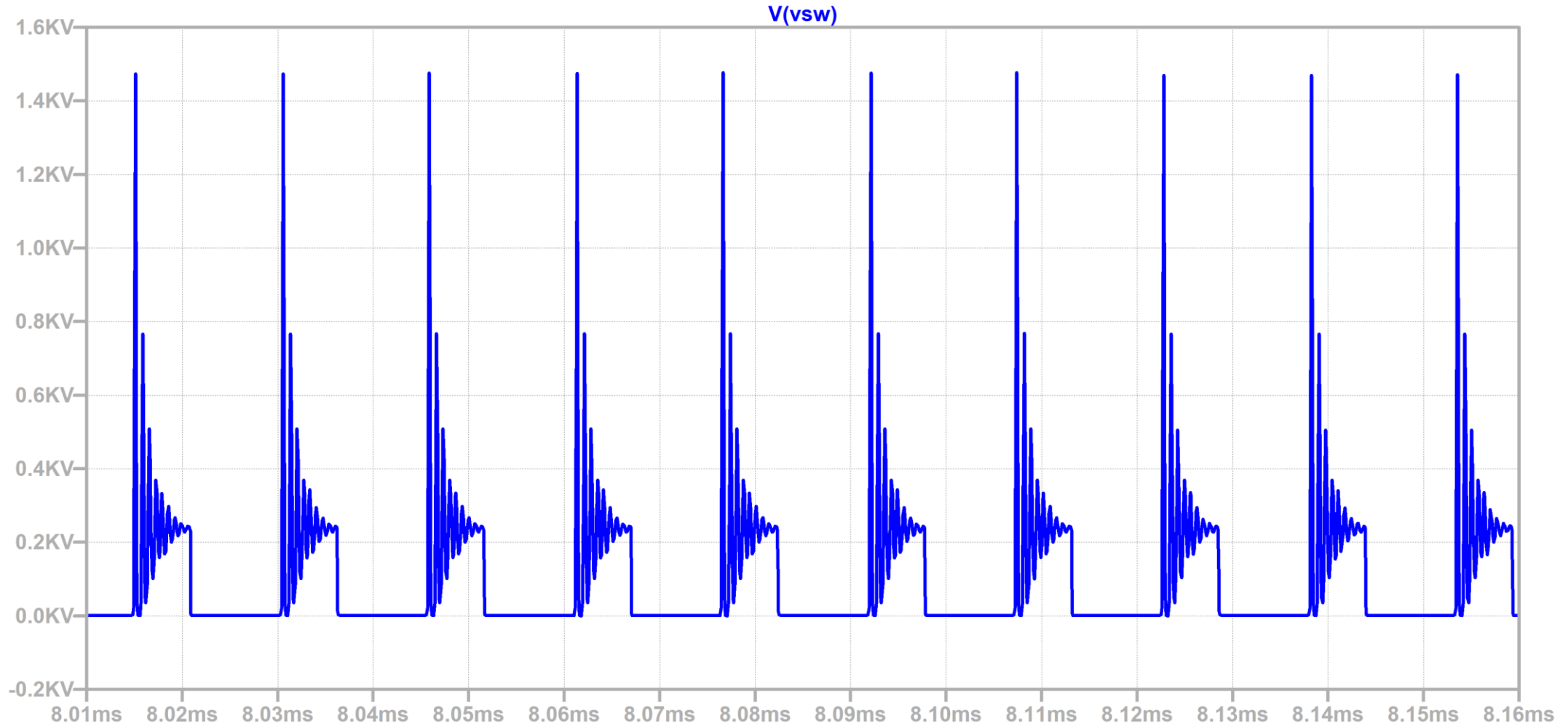
When the internal transistor in the DCDC controller (e.g. LTC3575) is turned off, the stray inductance tries to retain the current flowing to the ground causing a voltage increase at the node SW.



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Effect of Stray Inductance

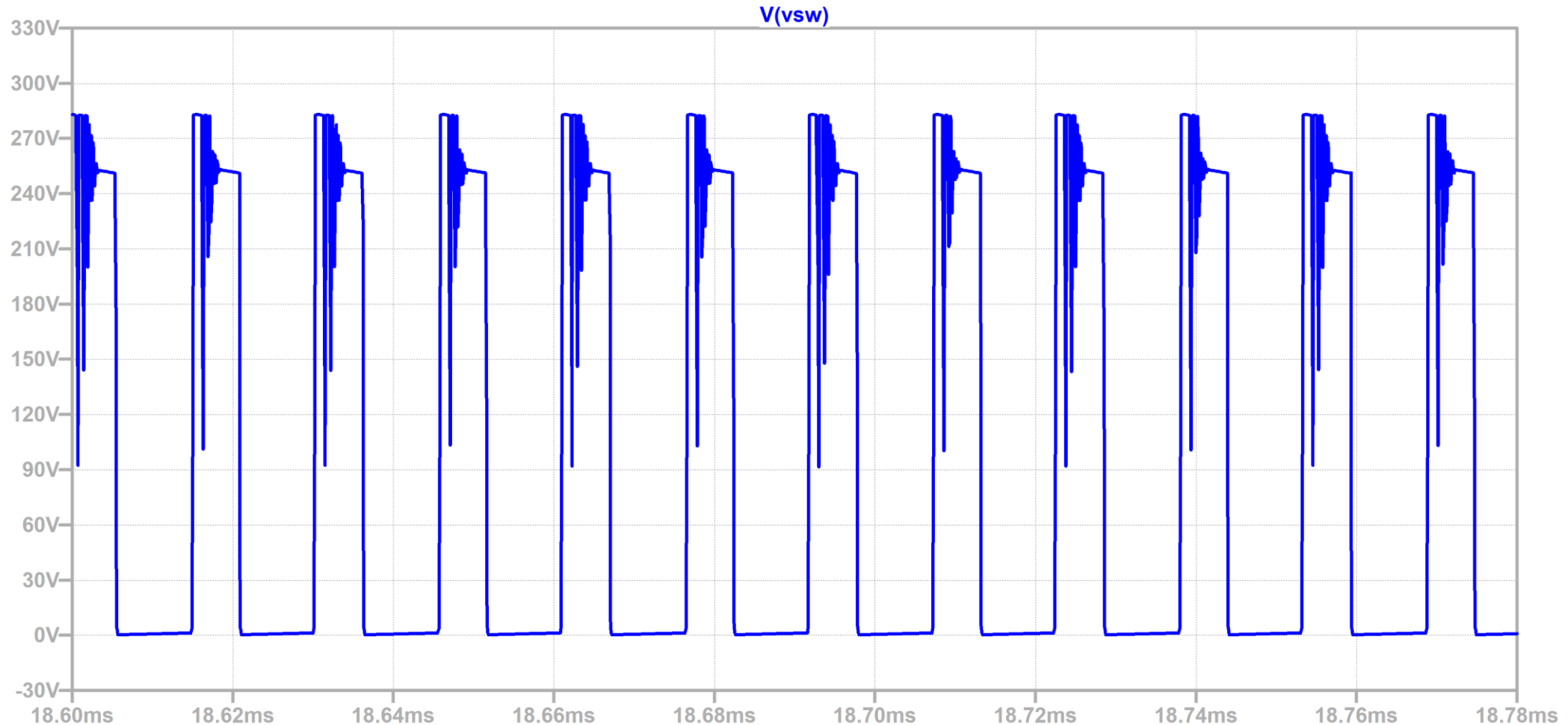
VSW voltage without Snubber Circuit: VSW exceeds 600 V, **M1** gets damaged



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

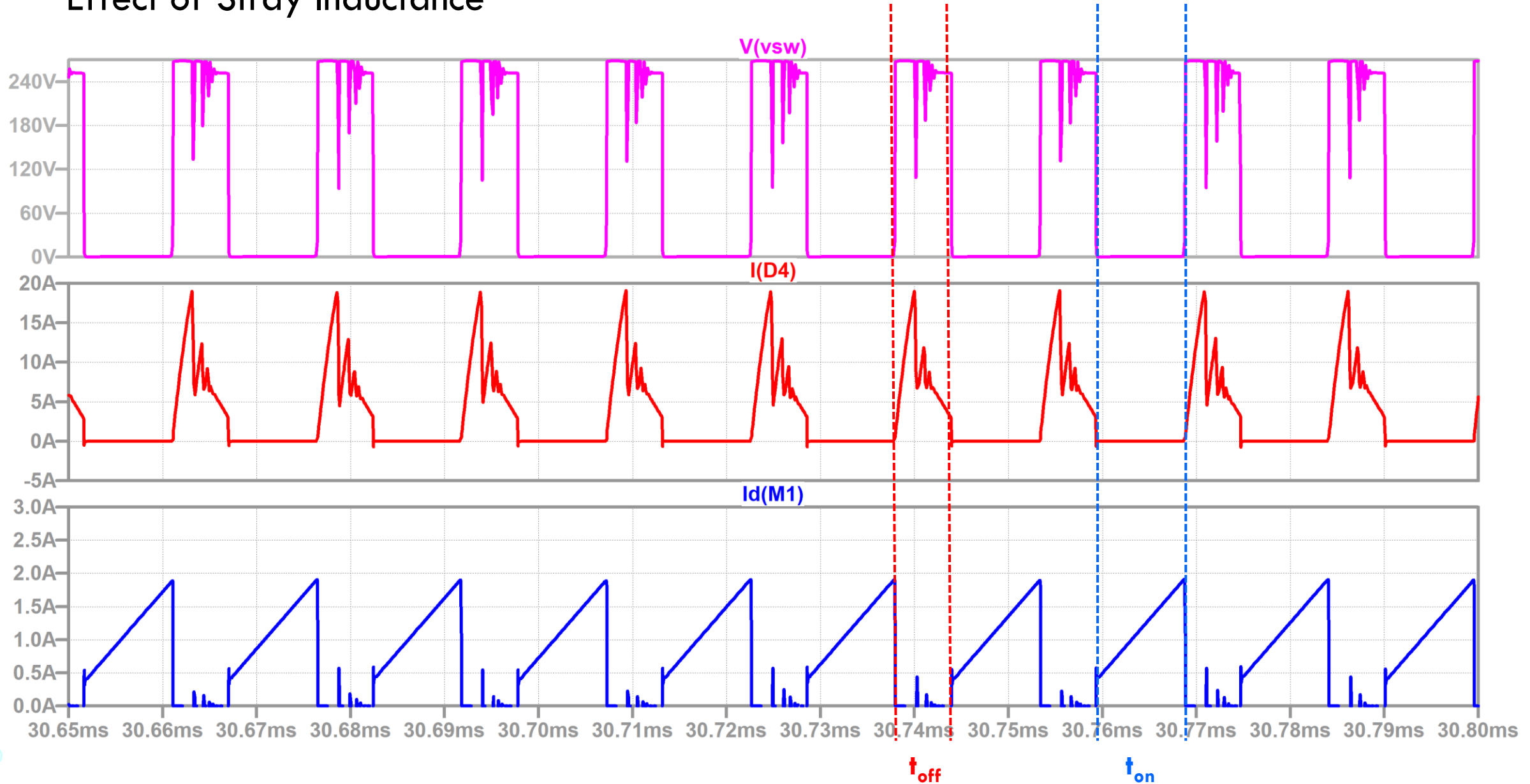
Effect of Stray Inductance

VSW voltage with Snubber Circuit: VSW remains below 600 V. There are still some oscillations which can be suppressed with further improvement of snubber circuit to improve EMC performance. However, a thermal analysis of snubber circuit has to be done to avoid damaging the snubber circuit.



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

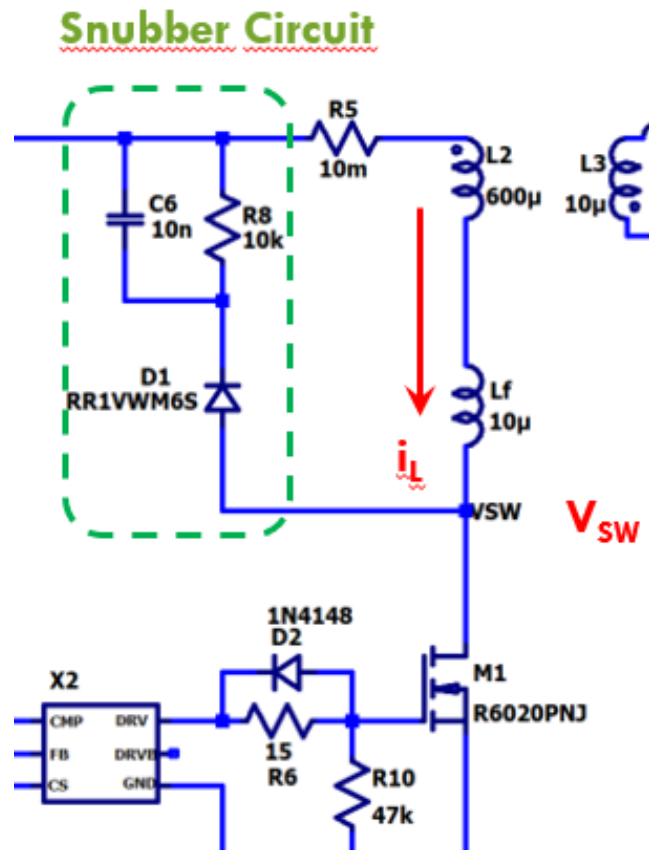
Effect of Stray Inductance



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Snubber Circuit Design - RCD

In some cases—especially in Flyback converters—the leakage-inductance spike can become so large that a RCD snubber circuit is necessary to clamp the voltage at V_{SW} .



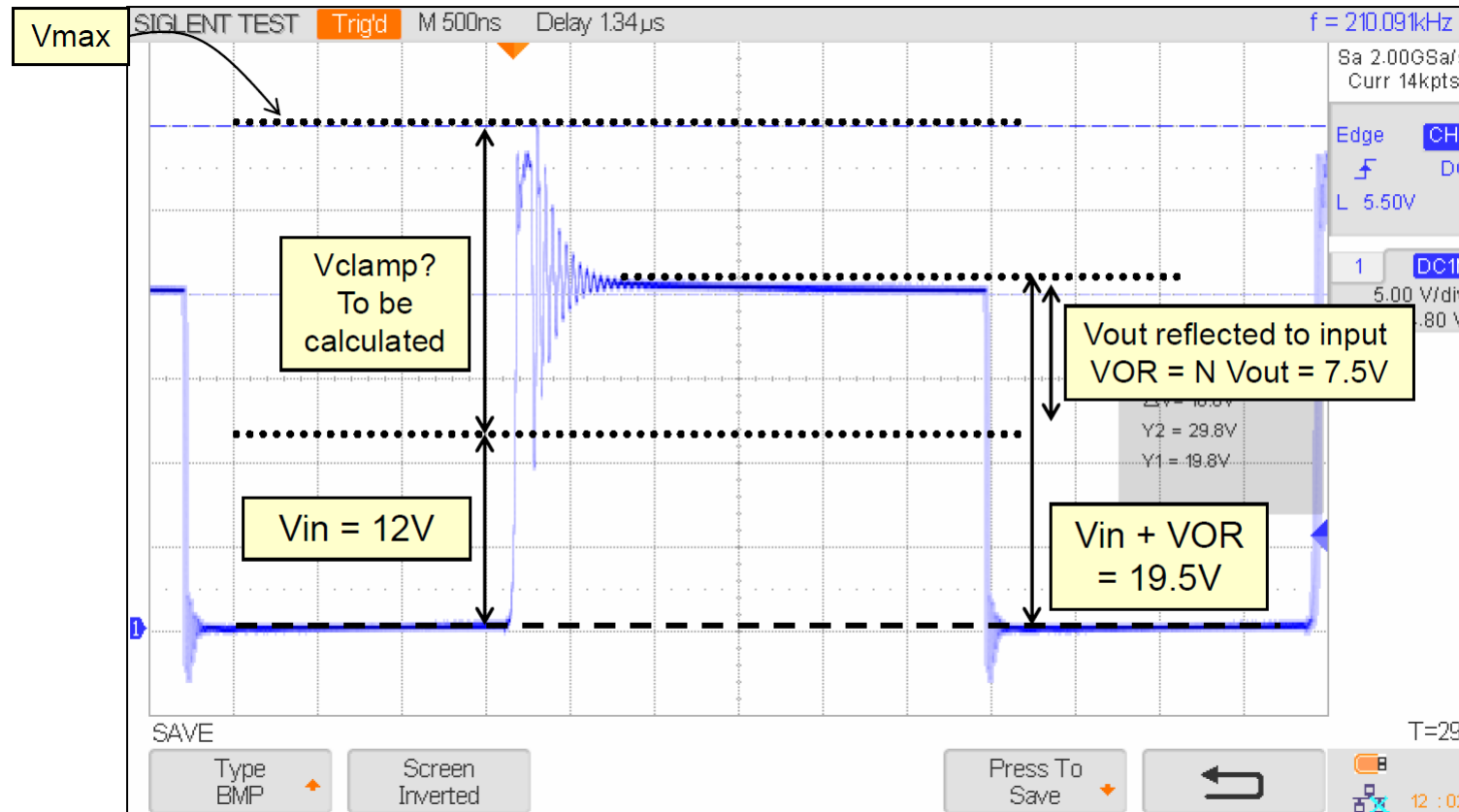
- We need a way to physically clamp or clip the spike to a safe voltage level so that the MOSFET is not overstressed or damaged.
- The most common solution used for this purpose is an RCD clamp circuit.

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Snubber Circuit Design - RCD

EMI: Electromagnetic Interference!!! → to be covered later

If max desired voltage on FET = 30V
Then $V_{\text{clamp}} = 30 - 12 = 18\text{V}$



FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Snubber Circuit Design - RCD

Step 1: Choose the maximum voltage ($V_{\max}$) allowed on the MOSFET. A higher allowable voltage reduces snubber losses, so we want to permit the highest safe spike without overstressing the MOSFET. As a rule of thumb, designers often select about 66% of the MOSFET's maximum rating to provide margin for overshoot. Alternatively, you can calculate the exact power you are willing to dissipate in the clamp network and then determine the allowable clamp voltage accordingly.

Step 2: Calculate V_{clamp} → from previous slide: $V_{\text{clamp}} = V_{\max} - V_{\text{in}}$

Step 3: * Calculate R_{clamp}

$$R_{\text{clamp}} = \frac{2 V_{\text{clamp}} (V_{\text{clamp}} - NV_{\text{out}})}{f_s L_{\text{leakage}} I_{\text{peak}}^2}$$

From Step 2

Vout reflected to input

From measurements, calculations or simulations
(given by transformer manufacturer)

* Christophe Basso – Switch-Mode Power Supplies: SPICE Simulations and Practical Designs

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Snubber Circuit Design - RCD

Step 4: Calculate C_{clamp} → the value of RCD capacitor does not impact the losses – Its value therefore is not crucial; it just needs to be large enough such that voltage remains constant during the snubber operation

$$C_{\text{clamp}} = \frac{5}{R_{\text{clamp}} F_s} \quad \text{based on} \quad RC \approx \frac{5}{F_s}$$

Step 5: Calculate the total snubber loss

$$P_{\text{loss_clamp}} = \frac{1}{2} L_{\text{leakage}} I_{\text{peak}}^2 F_s \left(\frac{V_{\text{clamp}}}{V_{\text{clamp}} - NV_{\text{out}}} \right)$$

Because of the inherently complex behavior of flyback converters, snubber-circuit calculations are never fully reliable. The analytical formulas provide only an initial estimate for component values. The final snubber design must always be verified and adjusted through laboratory measurements, including oscilloscope waveform analysis and thermal evaluation under real operating conditions.

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Snubber Circuit Design – RCD - Example

$$V_{in} = 12V, V_{out} = 7.5V, N = 1:1 \rightarrow N V_{out} = (1:1) \times 7.5V = 7.5V$$

$$I_{peak} = 2.5A, L_{leakage} = 250 \text{ nH}, F_s = 200 \text{ kHz}$$

$$\text{We select } V_{max} = 30V; \text{ therefore } V_{clamp} = 30V - 12V = 18V$$

$$R_{clamp} = \frac{2 V_{clamp} (V_{clamp} - N V_{out})}{F_s L_{leakage} I_{peak}^2} \Rightarrow R_{clamp} = \frac{2 \cdot 18 (18 - 1 \cdot 7.5)}{200,000 \cdot 250 \times 10^{-9} \cdot (2.5)^2} = 1200 \Omega$$

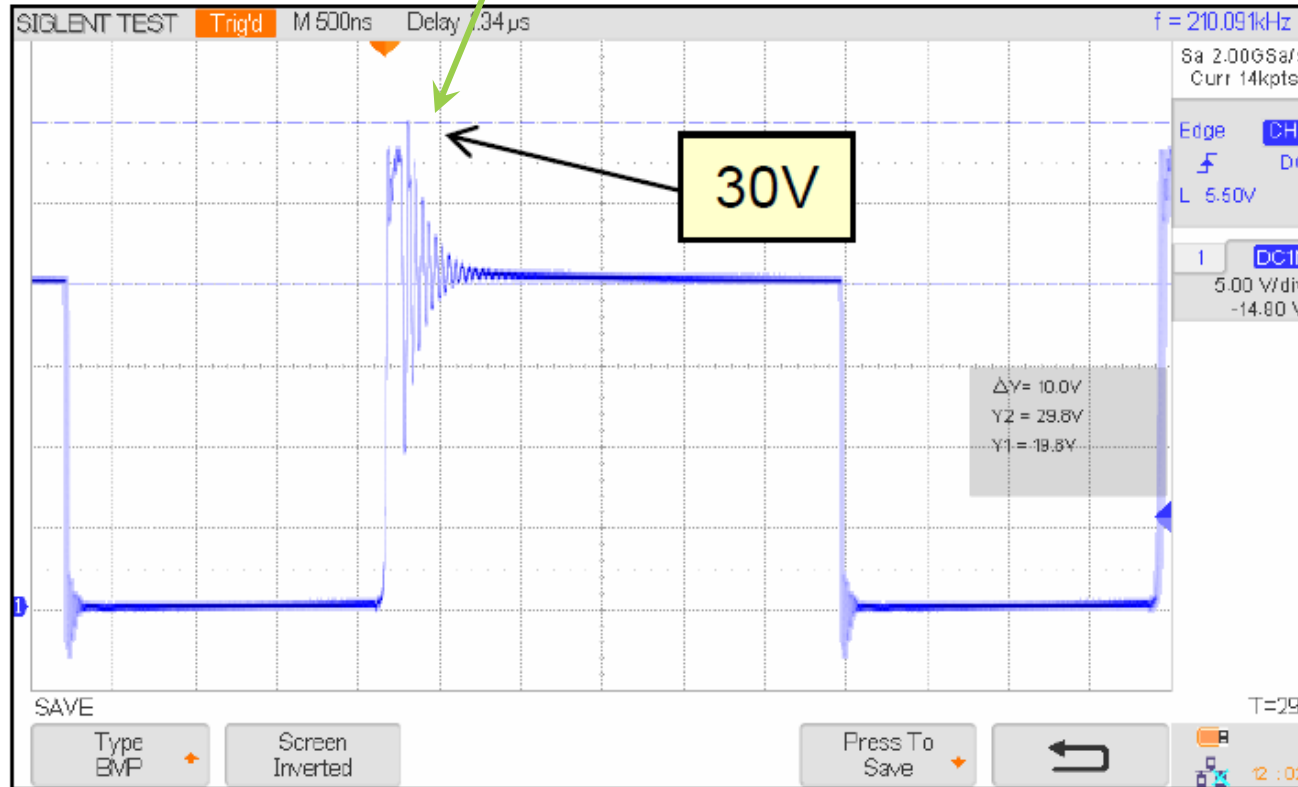
$$C_{clamp} = \frac{2.5 \text{ to } 5}{R_{clamp} F_s} \Rightarrow C_{clamp} = \frac{2.5 \text{ to } 5}{1200 \Omega \times 200,000 \text{ Hz}} = 10 \text{ to } 20 \text{ nF}$$

$$P_{loss_clamp} = \frac{1}{2} (250 \text{ nH}) (2.5 \text{ A})^2 (200 \text{ kHz}) \left(\frac{18}{18 - (1 \cdot 7.5)} \right)$$

FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE

Snubber Circuit Design – RCD - Example

Clamped to 30 V!



IMPORTANT METRICS OF DCDC CONVERTERS

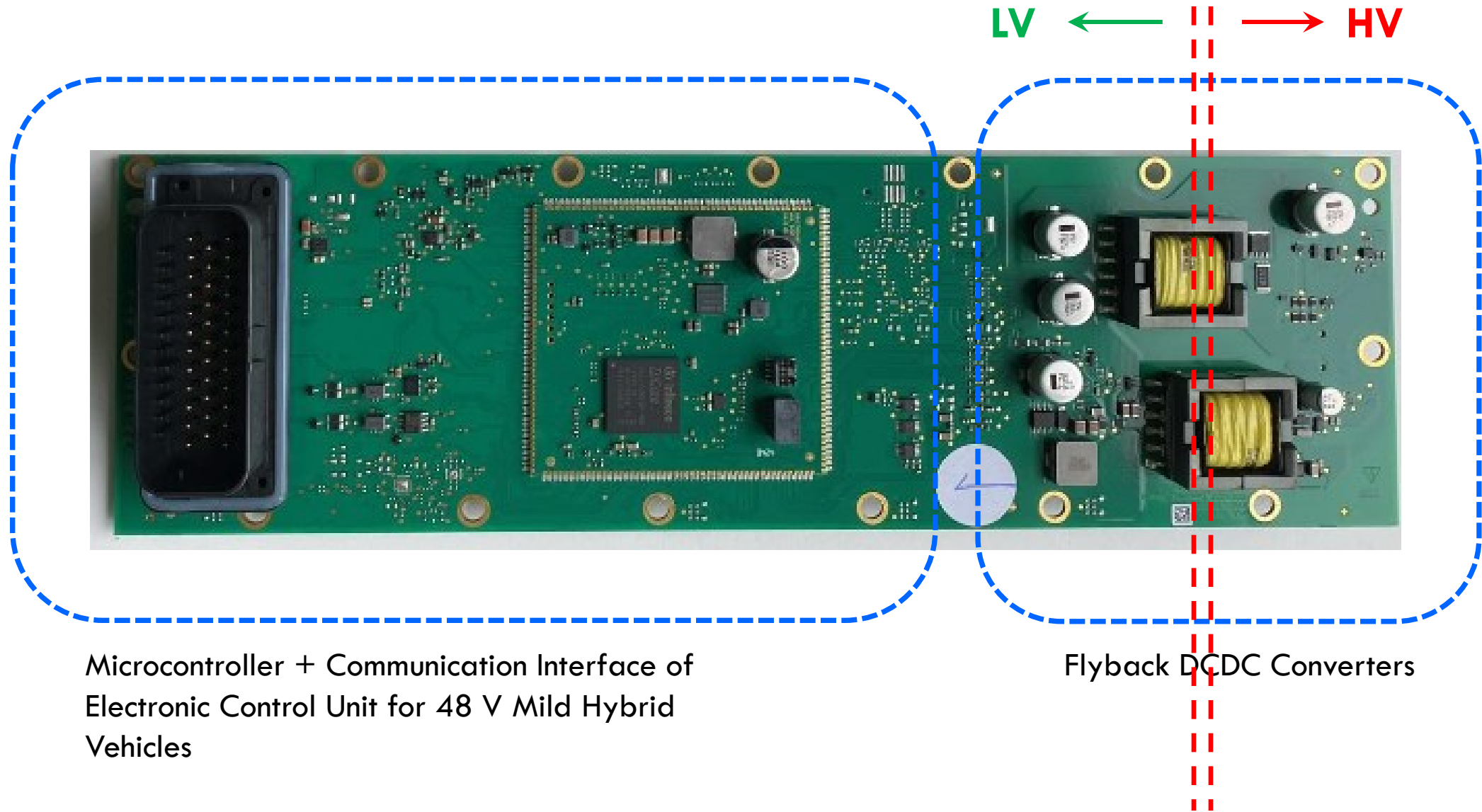
Important Metrics of DCDC Converters:

- Output voltage ripple
- Load regulation
- Input line regulation
- Maximum output power → power loss analysis of transistor, diodes, resistors and transformer
- Stability

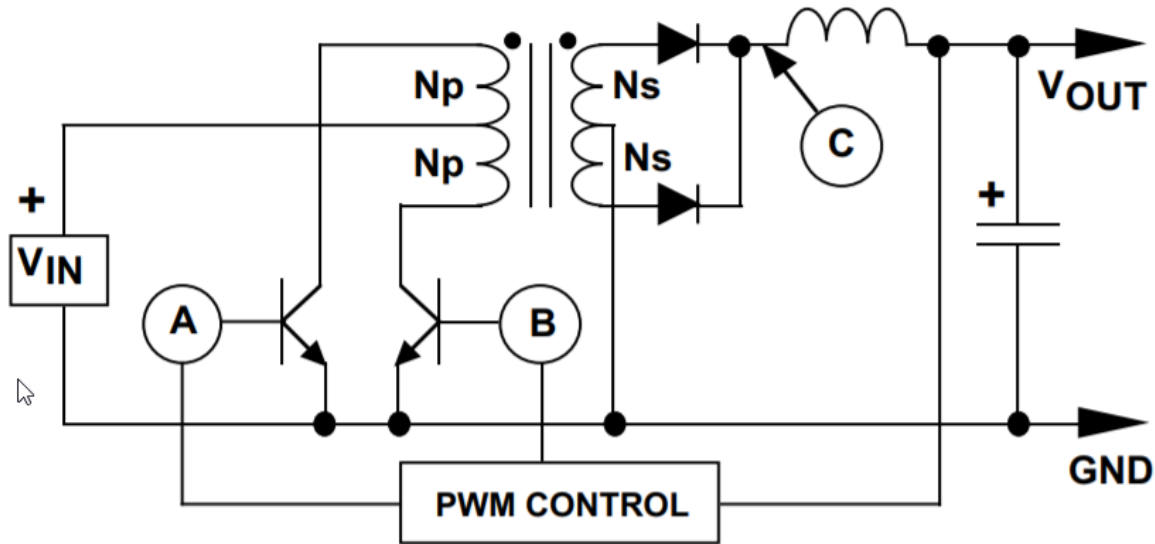
Output Voltage [V]	Line Regulation	Load Regulation	Output Voltage Ripple [mV]
12	0.5%	0.3%	100



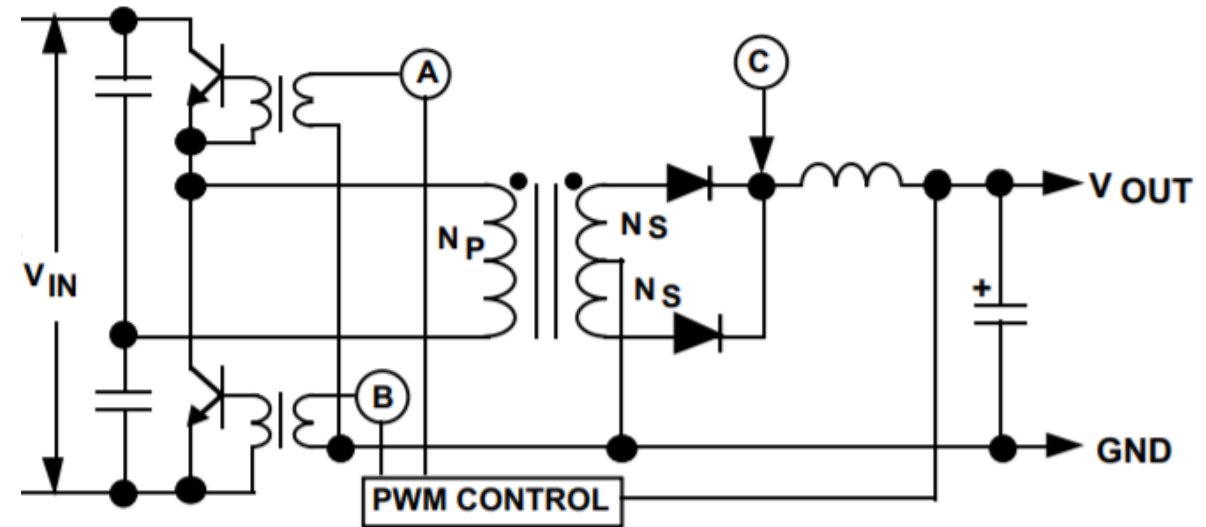
FLYBACK DCDC CONVERTER – APPLICATION EXAMPLE



OTHER ISOLATED DCDC CONVERTERS TYPES

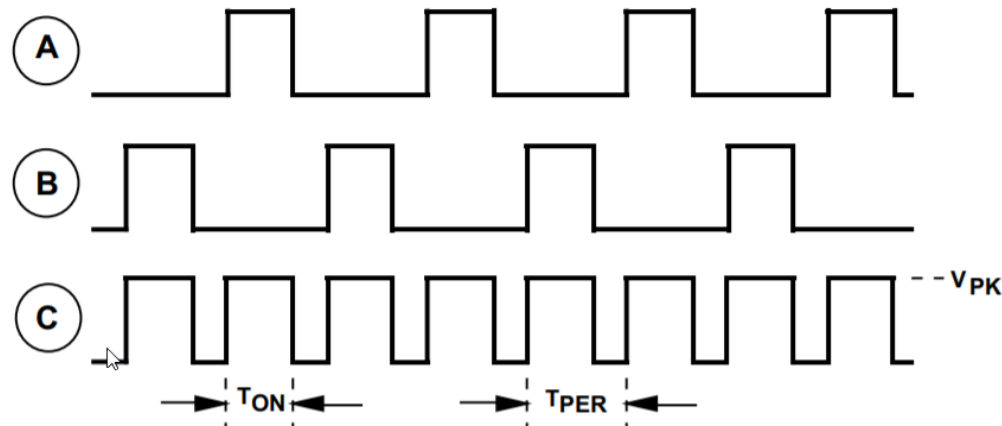


Push-Pull DCDC Converter



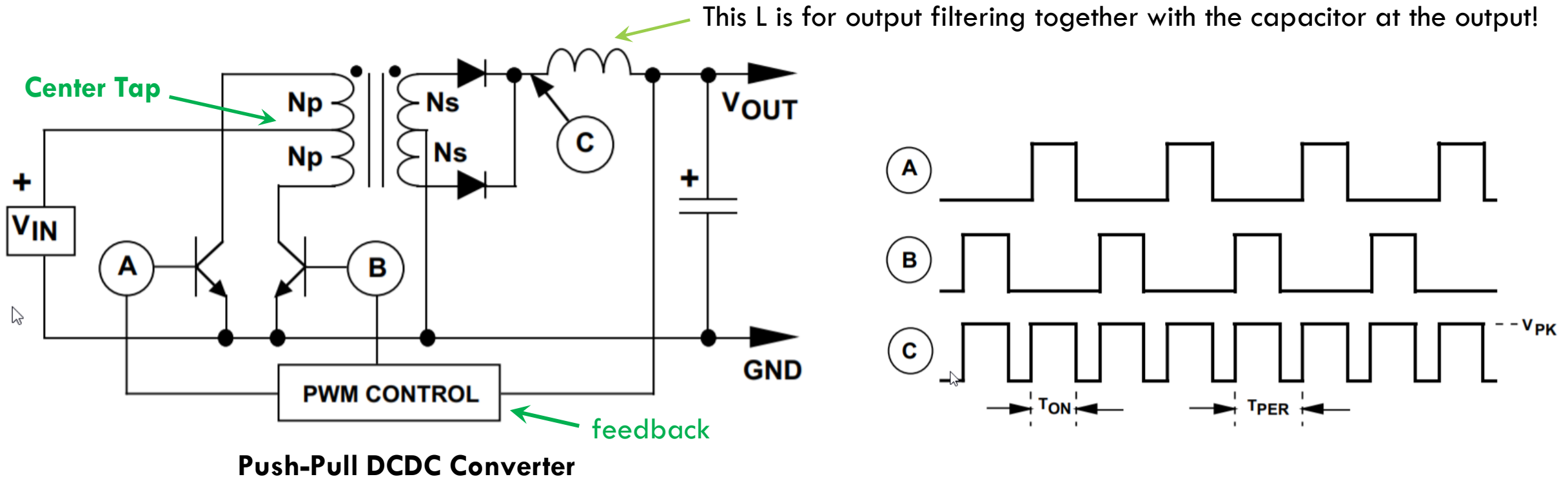
Half-Bridge DCDC Converter

Transformer secondary current flows at the same time as primary current (when either of the switches is on).



Half-Bridge and Push-Pull DCDC Converters will not be covered in detail within the scope of this lecture

OTHER ISOLATED DCDC CONVERTERS



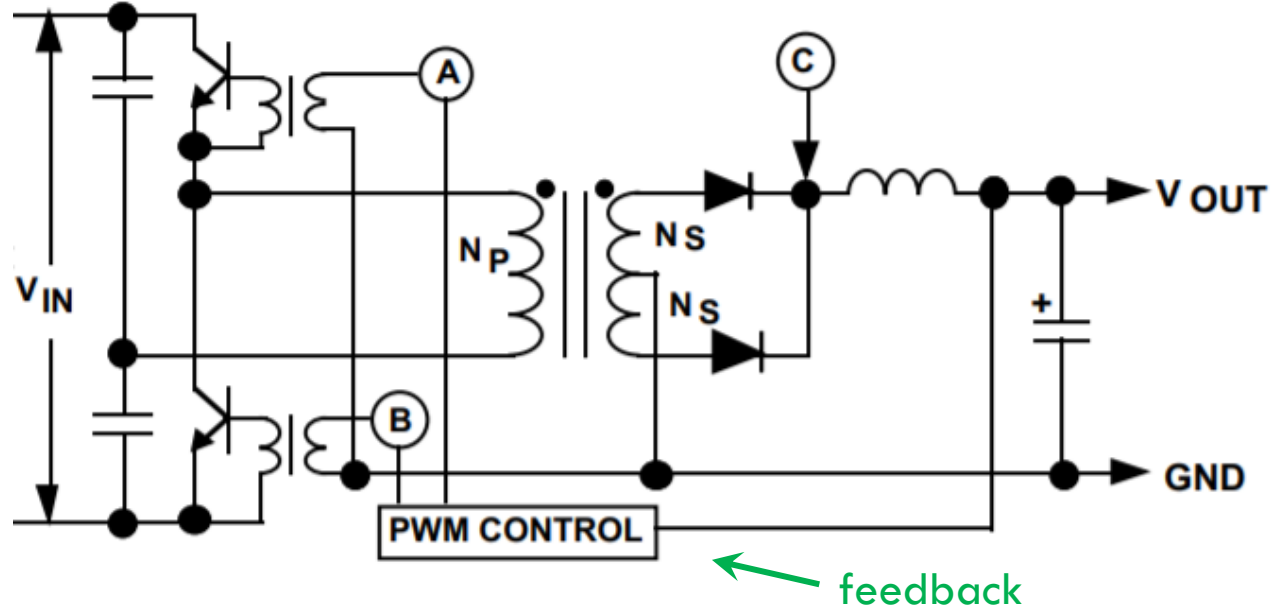
$$V_{OUT} = V_{PK} \times (T_{ON} / T_{PER})$$

The peak amplitude of the secondary pulses (V_{PK}) is given by

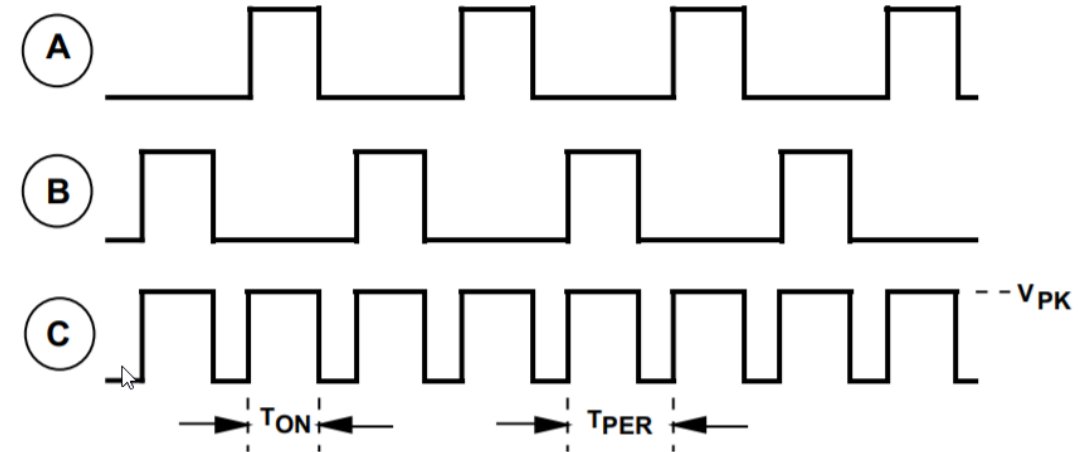
$$V_{PK} = (V_{IN} - V_{SWITCH}) \times (N_S / N_P) - V_{RECT}$$

Transformer secondary current flows at the same time as primary current (when either of the switches is on).

OTHER ISOLATED DCDC CONVERTERS



Half-Bridge DCDC Converter



$$V_{OUT} = V_{PK} \times (T_{ON} / T_{PER})$$

The peak amplitude of the secondary pulses (V_{PK}) is given by

$$V_{PK} = (1/2 V_{IN} - V_{SWITCH}) \times (N_S / N_P) - V_{RECT}$$

Transformer secondary current flows at the same time as primary current (when either of the switches is on).

OTHER ISOLATED DCDC CONVERTERS

Criterion	Flyback	Push-pull	Half-bridge
Energy transfer	Store then transfer	Direct transfer	Direct transfer
Magnetic element	Gapped transformer	Center-tapped transformer	Non-center-tapped transformer
Number of switches	1	2	2
Input voltage utilization	Full V_{in}	Full V_{in} (alternating)	$V_{in} / 2$
Switch voltage stress	High ($\approx V_{in} + \text{reflected}$)	High ($\approx 2 \cdot V_{in}$)	Moderate ($\approx V_{in}$)
Leakage inductance impact	High, snubber required	Medium	Lower
EMI	High	Medium	Lower
Efficiency	Low–medium	Medium–high	High
Typical power range	<150–200 W	~100–500 W	>200 W



SUMMARY

Flyback, push-pull, and half-bridge DC-DC converters differ fundamentally in how energy is transferred from the input to the output, and this directly determines the voltage and current stress experienced by their switching devices.

In a flyback converter, energy is first stored in the magnetizing inductance of the transformer while the primary switch is on and then released to the secondary when the switch turns off. Because of this energy-storage operation, the switch must block not only the input voltage but also the reflected output voltage during turn-off. In addition, leakage inductance causes voltage overshoot, further increasing the drain-source voltage stress.



SUMMARY

In a push-pull converter, energy is transferred directly to the secondary, but the primary winding is center-tapped and driven alternately by two switches. When one switch is conducting, the other switch must block approximately twice the input voltage due to the transformer action and polarity reversal. This leads to relatively high voltage stress on the switches. Moreover, any imbalance in duty cycle, timing, or device characteristics can introduce a DC flux component in the transformer, increasing peak currents and further stressing the switches.

In a half-bridge converter, the input voltage is divided by two series DC-link capacitors, and the transformer is driven from their midpoint. As a result, each switch only blocks approximately the input voltage when it is off, leading to lower voltage stress compared to flyback and push-pull topologies. Energy is transferred directly from the primary to the secondary without intentional energy storage in the magnetic core.

SUMMARY

From an efficiency perspective, these differences are decisive. Flyback converters suffer from high conduction losses due to large peak currents and high switching losses due to elevated voltage stress and snubber dissipation. Push-pull converters improve efficiency by avoiding energy storage but still experience higher voltage stress and additional losses caused by current imbalance and transformer saturation risk. Half-bridge converters typically achieve the highest efficiency because they combine lower switch voltage stress, lower RMS current, reduced leakage-related losses, As a result, half-bridge topologies are preferred in medium- to high-power applications where efficiency and reliability are critical.

In a half-bridge converter, lower switching voltage improves efficiency because switching losses are directly proportional to the voltage across the switch during turn-on and turn-off. Since each MOSFET in a half-bridge blocks only the input voltage (rather than twice the input voltage in push-pull or the input plus reflected output voltage in flyback), the voltage–current overlap during switching transitions is reduced. This lowers switching loss, reduces dv/dt and EMI, and allows the use of lower-voltage MOSFETs with lower $R_{DS(on)}$, all of which contribute to higher overall efficiency.

SUMMARY

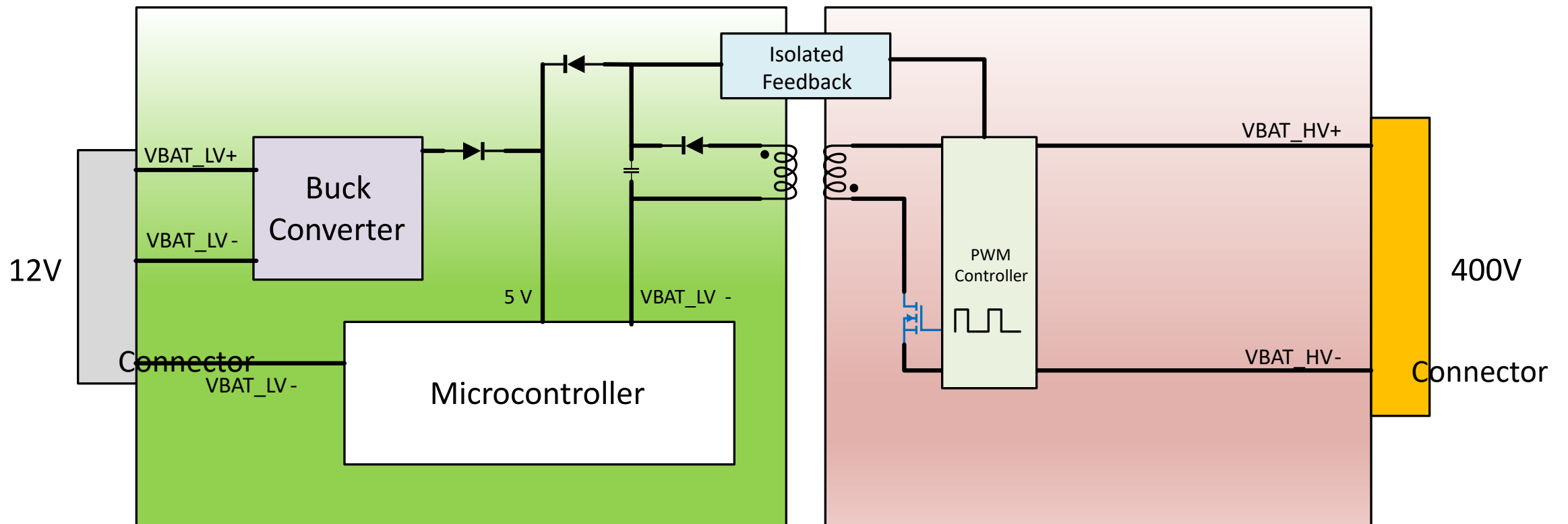
In a flyback converter, current conduction occurs only during one part of the switching cycle. The primary switch carries current only during the on-time, while no current flows in the primary during the off-time. Because the same amount of energy must be delivered in a shorter conduction interval (effectively a half-cycle), the current must rise to a higher peak value. In half-bridge and push-pull converters, current flows during both half-cycles through alternating switches, spreading the energy transfer over a longer time and therefore reducing the required peak current.



APPLICATION EXAMPLE – DCDC DESIGN

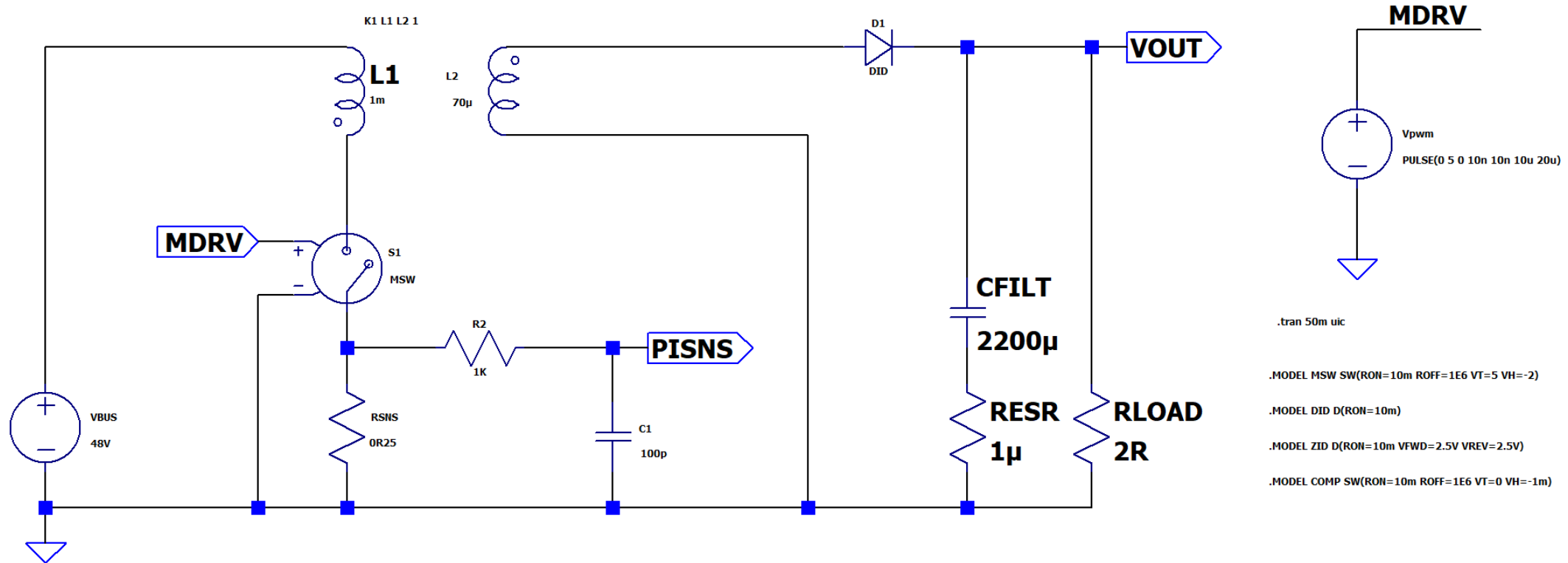
Simple HW Requirements of ECU Power Supply Concept

- The ECU shall be supplied by a 12V battery (low voltage).
- The ECU shall be supplied by a 400 V battery (high voltage).
- The HV and LV shall be galvanically isolated.
- The ECU supply voltage is 5.0 V.
- The ECU current consumption shall not exceed 200 mA.
- The power supply concept is shown in the figure below:



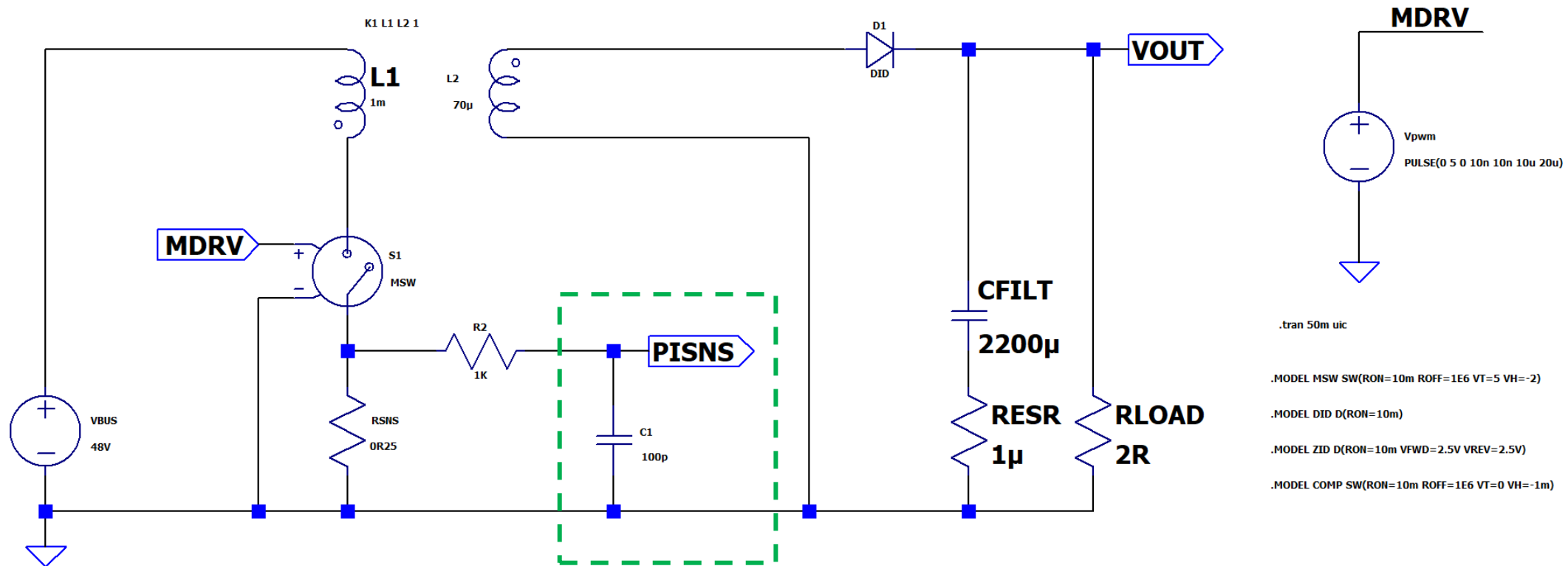
APPLICATION EXAMPLE – DCDC DESIGN

- A Flyback converter which converts 400 V down to 5 V. For simplicity, the feedback of the Flyback shall be neglected



APPLICATION EXAMPLE – DCDC DESIGN

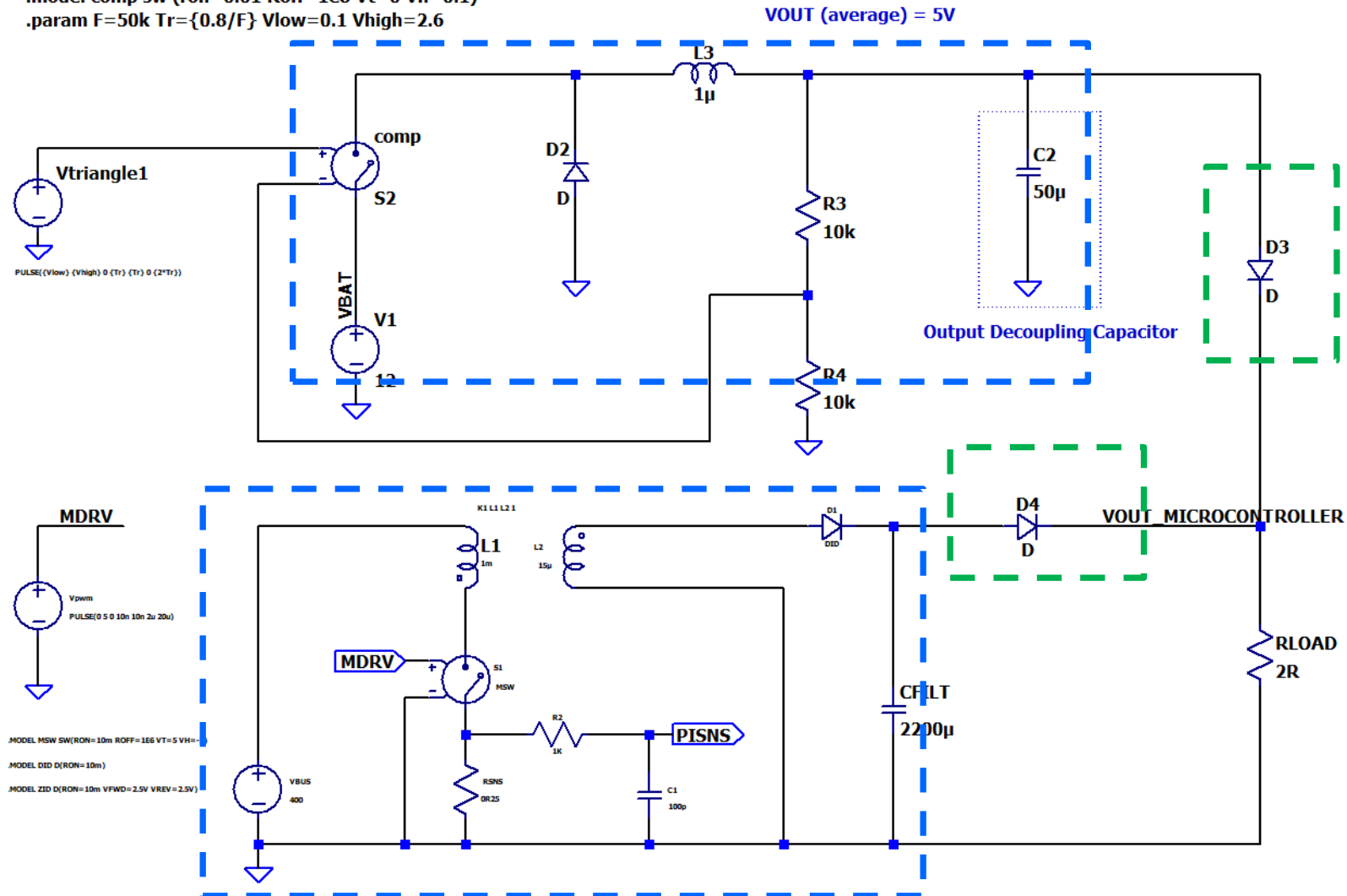
- A Flyback converter which converts 400 V down to 5 V. For simplicity, the feedback of the Flyback shall be neglected



Current Sense: Current can be monitored by a DCDC Controller for current control $\rightarrow I_{pri} = V_{PISNS} / 250m\Omega$

```
.tran 50m uic
```

```
.model comp sw (ron=0.01 Roff=1e8 Vt=0 Vh=0.1)
.param F=50k Tr={0.8/F} Vlow=0.1 Vhigh=2.6
```



References

- Ribbens, William B. *Understanding automotive electronics* / William B. Ribbens; with contributions to previous editions by Norman P. Mansour ... [et al.] – 6th ed..
- Analysis and Design of Analog Integrated Circuits”, Gray Mayer et al, 5th Edition, Wiley, 2009
- Würth Elektronik Flyback Transformer Application Note, 2018
- How to Design a Flyback Converter in Seven Steps, MPS Application Note
- Power Topologies Handbook, Markus Zehendner, Matthias Ulmann, Texas Instruments, 2020